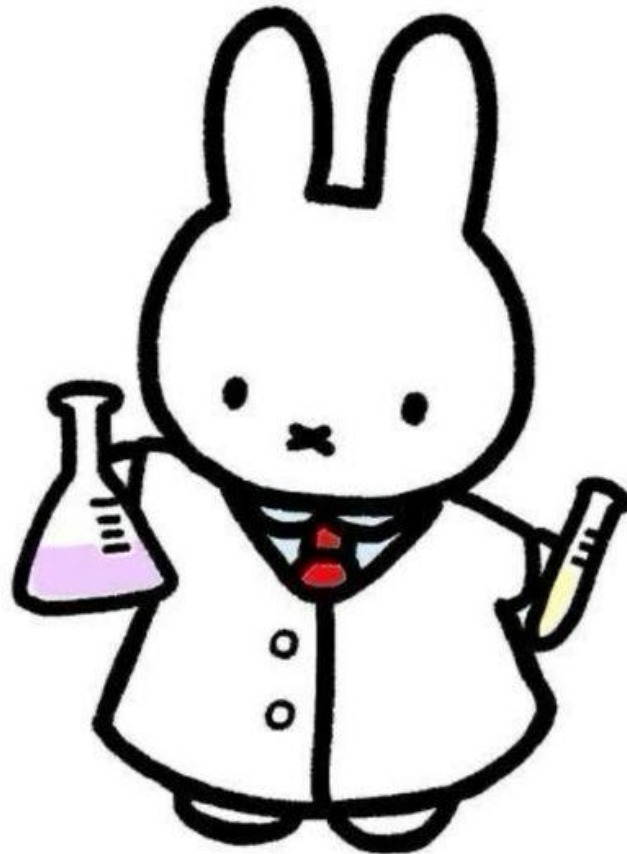


HSC BIOLOGY SYLLABUS SUMMARY NOTES



MODULE 5: HEREDITARY



Reproduction

Inquiry question: How does reproduction ensure the continuity of a species?

- explain the mechanisms of reproduction that ensure the continuity of a species, by analysing sexual and asexual methods of reproduction in a variety of organisms:

– **animals: advantages of external and internal fertilisation**

External fertilisation

Advantages	Disadvantages
– greater genetic variation – less intraspecific competition – wider ecological niche – greater chance of continuity – less energy cost	– gametes can easily die in unfavourable conditions – gametes are prone to predators (less protection)

e.g. Salmon breeds according to temperature cycles: climate change affects the breeding cycle. The gametes cannot live in the wrong temperature.

Coral: climate change affects ability for polyps to live and reproduce

Internal fertilisation

Advantages	Disadvantages
– Can move around in regions due to the zygote being produced within the reproductive tract – Increased fertilisation	– Less genetic variation -> due to parents choosing the other – High energy cost-> in growing offspring and raising/taking care of them – Disease e.g. STD especially if there is zero immunity – Slow population growth rate (compared to external fertilisation) – Less offspring produced

– **plants: asexual and sexual reproduction**

- Cross- pollination: the transfer of pollen from one plant's anther to the stigma of another plant within the same species. Increases genetic variation in offspring
 - Plants that pollinate via wind -> have larger reproductive systems and larger SA
 - Plants that pollinate via biological vector-> scented, colourful, aesthetically shaped
- Seed Dispersal:
 - Plants that grow by rivers and streams: seed shell, cotton type structures within for increased buoyancy
 - Tall flowering plants: balanced wings to allow for dispersion through the air
 - Drifters: flimsy weightless additions
 - Plants that attract biological vectors: seeds that have sharp spines, hooks
- Angiosperms-> flowering plants: **Monocots and dicots**
 - -> -> Pollen is transferred to the stigma of a flower
 - > the pollen's tube cell creates a pollen tube from the stigma towards the ovary
 - > the pollens generative cell travels down the tube and divides to form two sperm
 - > the two sperm enter one ovule
 - > one sperm fertilises the egg that other sperm combines with the polar nuclei to form the endosperm to provide nutrients to the zygote
 - > the ovule matures into a seed
 - > the zygote develops into an embryo which grows into a new plant after the seed germinates.
 - **Polar nuclei**-> develops from the other ova in other ovaries combined to form a diploid. The second sperm fuses with the diploid cell to create the endosperm. The endosperm removes/limits chance of double fertilisation and acts as source of nutrient for the zygote during early stages of growth
- Gymnosperms: e.g. pines, cycads and ginkgoes.

- Sporophyte (male cone) has spore sacs which contain male gametes. The mature sporophyte lands on an ovule (female eggs on leaves). The male gametes fuse with the female gametes, producing offspring.

- **fungi:** budding, spores
- **bacteria:** binary fission
- **protists:** binary fission, budding

Types of asexual reproduction

Budding	A new organism develops from an outgrowth, or a bud of another organism	Fungi, protists, plants e.g. yeast, hydra and plants
Spore formation	Produced in great numbers by fungi and some plants by structures called sporangia. Spores are lightweight to be easily distributed.	Fungi, some plants e.g. mushrooms, mosses, ferns
Binary Fission	When one cell splits into two identical daughter cells. Very efficient. Types: transverse, longitudinal, irregular and oblique	Bacteria, fungi, protists e.g. amoeba, E.coli
Vegetative reproduction/propagation	Some plants and fungi produce bulbs or organs that can be cut off and regrown	Plants, fungi e.g. tissue culturing
Fragmentation of body parts	Occurs when a body part breaks off the parent and a new body part form's	Animals e.g. starfish, flatworms

- analyse the features of fertilisation, implantation and hormonal control of pregnancy and birth in mammals

Follicle	A small sac found in the ovary that contains one immature egg cell. As this structure grows and matures, it releases oestrogen.
Corpus luteum	This structure develops after follicle ruptures and releases an egg during ovulation. It releases oestrogen and progesterone.
Endometrium	This tissue lining of the mammalian uterus. A fertilised egg implants into this lining.
Oestrogen	A hormone produced by either a mature follicle or the corpus luteum. It stimulates the endometrium to thicken.
Progesterone	Hormone produced by corpus luteum to maintain the endometrium.
FSH	Hormone that stimulates the growth and development of ovarian follicles
LH	A hormone that triggers ovulation and the formation of the corpus luteum
Follicular phase	The first half of the ovarian cycle characterised by one follicle maturing. Oestrogen levels rise during this phase.
Luteal phase	The second half of the ovarian cycle characterised by the corpus luteum producing oestrogen and progesterone, and eventually degeneration if follicle is not fertilised.

- evaluate the impact of scientific knowledge on the manipulation of plant and animal reproduction in agriculture

Selective breeding-> the creation of organism with certain desirable characteristics

Natural breeding	Artificial insemination	Artificial pollination	Cloning
Placing male and female in an enclosed environment and waiting	Taking sperm from a male animal and inserting it directly into the female's reproductive tract	Taking pollen from one flower and inserting it into another flower using a small brush	Creating a genetically identical copy of an organism

Cell Replication

Inquiry Question: How important is it for genetic material to be replicated exactly?

- model the processes involved in cell replication, including but not limited to:

– **mitosis and meiosis**

Mitosis:

- **Interphase:** DNA replicates.
- **Prophase:** the chromosomes condense to become visible and the nuclear envelope breaks down. The mitotic spindle begins to form
- **Metaphase:** Chromosomes align in the middle of the cell and each centromere of a chromosome becomes attached to a spindle fibre
- **Anaphase:** The two sister chromatids of each chromosome move away from each other towards opposite ends of the cell.
- **Telophase:** The cell prepares to split in two by forming a nuclear envelope around each set of chromosomes and the spindle disappears.
- **Cytokinesis:** divides cytoplasm into two, each with their own nucleus.

Meiosis:

- **Prophase 1:** DNA condenses, nuclear envelope disintegrates. Synapsis occurs (Chromosome separate into homologous pairs- tetrads). Crossing over occurs.
- **Metaphase 1:** Tetrads line up in the middle. Independent assortment-> so various combinations of paternal and maternal chromosome alignment are possible-> Random segregation
- **Anaphase 1:** Tetrads are split apart. No homologous pairs
- **Telophase 1:** Nuclear membrane form, two daughter cells form.
- **Cytokinesis 1:** Daughter cells are not identical to each other and have half the original number of chromosomes
- **Metaphase 2:** Chromosomes align in the middle
- **Anaphase 2:** Daughter chromosomes move apart to opposite poles.
- **Cytokinesis 2:** Four resulting daughter cells are not genetically identical to each other and have half the original chromosome number.

– **DNA replication using the Watson and Crick DNA model, including nucleotide composition, pairing and bonding**

1. **Unravelling DNA:** Topoisomerases unwind ds DNA strands and relaxes them to prevent super coiling. Helicase breaks the hydrogen bonds between bases “unzipping” the double helix into two single strands.
2. **Building a new strand:** semi-conservative-> when DNA is produced, one of the strands in each new DNA molecule comes from the old DNA molecule. DNA polymerase attaches free floating nucleotides to their complementary bases on the exposed strands. However, nucleotides only attach in a 5 prime to 3 prime directions. The leading strand is the 5 prime to 3 prime strand and attaches bases without issues. The lagging strand causes bases to attach the opposite way causing fragments called Okazaki fragments. DNA ligase glues the fragments together.
3. **Forming the DNA backbone:** DNA polymerase forms the sugar-phosphate backbone of the new DNA strand. Ligase creates hydrogen bonds and gyrase winds up the strands.

- assess the effect of the cell replication processes on the continuity of species

Mitosis-> exact replication, 2 daughter cells

- Positive effect: growth, tissue and cell repair, genetic stability and functioning of the body.
- Negative effect: Incorrect DNA replication causing rapid cell growth (cancer), mutations, and possibility for incorrect protein functionality

Mitosis-> crossing over, independent assortment, random segregation, half and half chromosomes from parents, 4 daughter cells

- Positive effect: efficiency of fertilisation, can overcome challenges in external environment using survival of the fittest and evolution by natural selection.
- Negative effect: missing chromosomes result in genetic disorder, and is passed onto subsequent generations

DNA and Polypeptide Synthesis

Inquiry question: Why is polypeptide synthesis important?

- Model the process of polypeptide synthesis, including:

- **Transcription and translation**

Transcription: RNA polymerases bind to a part of the DNA called the promoter region and the DNA unzips only the part of the DNA that contains the gene code needed. One strand is called the template strand, in which the RNA nucleotides code against, and the other strand is the coding strand (as it has the same code as the mRNA molecule being coded). The RNA nucleotides code against the template strand, forming a complementary single-stranded mRNA molecule. However this is known as the pre-mRNA strand

RNA Processing: Pre-mRNA strands contain coding sequences of nucleotides called exons, and nucleotides called introns which do not code. Directly after transcription, mRNA is spliced, and introns are removed by a spliceosome. The mRNA strands leaves the nucleus.

Translation:

Initiation-> occurs when the ribosome move along the mRNA molecules attaching the tRNA molecules by temporarily pairing the bases of the tRNA anticodons with the complementary triplets of bases (codons) on the mRNA. First codon is always AUG (start codon).

Elongation-> More tRNA molecules code, 3 tRNA anti-codons at a time. The amino acids from the tail end of each tRNA are linked to one another by an enzyme to form a polypeptide chain. Each amino acid is then spliced off its tRNA carrier. The amino acid chain grow as more anti-codons attach to the mRNA.

Termination-> When a tRNA reaches a stop codon, no more tRNA anti-codons attach and the polypeptide chain is processed in the cell and folded to form a protein.

- **Assessing the importance of mRNA and tRNA in transcription and translation**

mRNA (*messenger RNA*)

- Single- stranded
- Functions as an intermediate molecule
- Carries information from the DNA in the nucleus to the ribosomes in the cytoplasm

tRNA (*transfer RNA*)

- Three unpaired bases called an anti-codon
- Attaches to the complementary bases (codons) of the mRNA strand
- The specific sequence of three bases at the anti-codon end determines which amino acid is being carried by that specific tRNA molecule

rRNA (*ribosomal RNA*)

- Forms a structural part of ribosomes and is made in the nucleolus

- **Analysing the function and importance of polypeptide synthesis**

- **Assessing how genes and environment affect phenotypic expression**

Polypeptide synthesis creates proteins that carry out different functions in the body. In specialised cells in multicellular organisms, only certain genes are expressed in each cell type. Coded instructions for the production of a particular protein (or group of proteins) are said to be 'switched on' in the DNA of that cell. This ensures that each cell develops the necessary structures. **e.g. skin tissue, genes for the pigment melanin and the protein keratin are switched on in each cell ensuring that the cells become skin cells.**

Factors that affect phenotypic expression

- ➔ **Genes:** Due to hereditary genes, you may be more inclined to receive a certain disease more than others, especially if there is history of that disease within your family e.g. eye colour, blood type
- ➔ **Environment:** Differences in phenotype caused by environmental factors are not passed from one generation to the next e.g. suntan, piercings

- Investigate the structure and function of proteins in living things

Proteins are made up of one or more long chains of nitrogen-containing amino acids.

Function:

- Structural proteins: growth, repair, maintenance of tissue e.g. collagen
- Enzyme: biological catalyst e.g. DNA polymerase
- Hormones: secreted into the blood by endocrine e.g. insulin
- Immunity: Antibodies are proteins involved in the immune response

Structure:

1. **Primary structure**-> the amino acids sequence
2. **Secondary structure**-> with folding as a result of hydrogen bonding either into a spiral (alpha helix) or a pleated sheet
3. **Tertiary structure**-> caused by interactions between the polypeptide and its immediate environment. Forces of attraction between alpha helices and pleated sheets causing the polypeptide to fold into a more complex 3D shape
4. **Quaternary structure**-> occurs in proteins that are made up of two or more polypeptide chains to create an even more complex 3D shape caused by relations between individual subunits of the amino acids.

Genetic Variation

Inquiry question: How can the genetic similarities and differences within and between species be compared?

- Conduct practical investigations to predict variations in the genotype of offspring by modelling meiosis, including the crossing over of homologous chromosomes, fertilisation and mutations
crossing over of homologous chromosomes: Results in the exchange of genes between chromosomes pairs in resulting different combinations of alleles, different to the parents.
Random fertilisation: Occurs during sexual reproduction. The gamete released each cycle is random, which is genetically diverse. The sperm is similarly unique. This increases genetic variation in sexual reproduction.
Random segregation: During meiosis, genes on different chromosomes sort independently. The line up in the middle and the spindles pull random chromatid.

- Model the formation of new combinations of genotypes produced during meiosis, including but not limited to:
 - Interpreting examples of autosomal, sex-linked, co-dominance, incomplete dominance and multiple alleles
 - Constructing and interpreting information and data from pedigrees and Punnett squares

Autosomal dominant	Has a dominant genotype-> can be heterozygous or homozygous	➔ The trait is shown in all generations ➔ Affected parents can have unaffected kids
Autosomal recessive	Has a recessive genotype-> only homozygous	➔ Skips a generation ➔ Both unaffected parents having affected kids
Sex-linked	Alleles that are carried on the X or Y chromosome	➔ Occurs more often in males

		→ Can skip generations → All sons of an affected female are also affected
Incomplete dominance	The phenotype of heterozygotes appear as a blend of the phenotypes of either type of homozygote	→ Heterozygous e.g. pink snapdragons chinchilla rabbits
Co-dominance	The phenotype of heterozygotes involves both alleles being expressed, with neither dominant over the other	→ Heterozygous e.g. roan cow

- Collect, record and present data to represent frequencies of characteristics in a population, in order to identify trends, patterns, relationships and limitations in data, for example

Population genetics -> the study of how the gene pool of a population changes over time, leading to a species evolving.

Gene pool -> the sum total of all the genes and alleles within a population

Gene flow -> transferring of alleles between populations of a species. Without gene flow (lack of necessity, environmental change, geographic isolation) divergent evolution occurs (drastic changes in gene pool)

Genetic drift -> change in allele frequency of a population due to random events such as invasive species, natural disasters, climate events.

– Examining frequency data

Allele frequency: Is a relative proportion of a particular allele in a population.

Allele frequency = $\text{Number of an allele in a population} \div \text{total number of alleles in a population}$

Frequency of allele G: measure of how common an allele is within a population

Frequency of allele G = $\text{number of copies of allele (G) in the population} \div \text{total number of copies of the gene (G + g) in the population}$

Hardy-Weinberg model -> predicts allele frequency in a non-evolving population.

$$P^2 + 2pq + q^2 = 1$$

P = frequency of dominant allele

Q = frequency of the recessive allele

– Analysing single nucleotide polymorphism (SNP)

An SNP is a typing error in DNA. SNPs arise during DNA replication, when a single nucleotide is inserted incorrectly (insertion mutation) creating an error in the DNA sequence.

- To be termed an SNP, this mutation must occur in a least 1% of the population.
- SNPs cause variation in organisms such as:
 - Phenotypic change
 - Enzyme functioning
 - Disease susceptibility
 - Response to drugs
- SNPs occur in the non-coding regions of DNA (introns)
- SNPs are important genetic markers that are used to distinguish individuals. *Genetic markers* -> identified sequence of DNA at a known site (locus) on a chromosome e.g. STR (introns that contain a string of repeated nucleotides e.g. AGAGAG)

SNPs are used for:

- Forensics (ancestry, identifying background)
- Genetic markers (allow for genotypes to be differentiated)
- Used for diagnosis, early detection, preventative treatment
- Cause responses to drugs (personalise medication, minimise allergic responses)

Inheritance Patterns in a Population

Inquiry question: Can population genetic patterns be predicted with any accuracy?

- Investigate the use of technologies to determine inheritance patterns in a population using, for example

- **DNA sequencing and profiling**

In DNA sequencing the exact nucleotide sequence of a gene on a chromosome is determined. It identifies difference and similarities. Modern computation allows patterns to be identified.

DNA profiling or DNA fingerprinting allows for the identification of an individual's DNA characteristic, by identifying STRs.

- **PCR (polymerase chain reaction):**

1. STR loci is targeted in a biological sample with primers and is amplified.
2. DNA fragments are separated by gel electrophoresis.
3. Fluorescent dye reveals DNA bands that can be used for comparison of samples

- **RFLP (restriction fragment length polymorphism):**

1. Collection of DNA samples from cells (swabs, blood, hair follicles)
2. DNA cut into smaller pieces using restriction enzyme. Creates a mixture of DNA fragments of differing sizes because they contain variable number of tandem repeats.
3. DNA fragments separated by gel electrophoresis.
4. Gel visualised to show band patterns. Each individual will have a unique band pattern due to their individual gene sequence
5. Patterns of bands can be compared.

Human genome project -> worlds largest collaborative biological project to sequence the human genome. Goal to provide a complete and accurate sequence of the 3 billion DNA base pairs that make up the human genome and to find all of the estimated 20,000 to 25,000 human genes.

Positives:

- Determine gene function
- Early detection and diagnosis of human disease
- Sequencing genes of many organisms

- Investigate the use of data analysis from a large-scale collaborative project to identify trends, patterns and relationships, for example:

- **The use of population genetics data in conservation management**

Principle aim is to avoid extinction of species by applying conservation methods that ensure maintenance of biodiversity.

- Relies on gathering genetic data and making informed decisions about protected endangered to nearing extinction species
- Method of DNA analysis to identify segments of the genome essential for organism adaptation in the environment.

- **Population genetics studies used to determine the inheritance of a disease or disorder**

Genetic testing can help diagnose disease and the probability of future offspring inheriting these conditions.

- Most common genetic difference are SNPs.
- NSW newborn screening program provides potential for large-scale genomic analysis of newborns.

- **Population genetics relating to human evolution**

Human evolution:

Anthropological genetics -> determine pathways of human evolution. Mathematical population models allow scientists to interpret genetic diversity and predict the genetic patterns associated with human evolution.

Human migration theories:

Multiregional hypothesis: suggests all human populations can be traced back to when homo erectus first left African 2 million years ago.

- Suggests gene flow between neighbouring populations
- Relies mostly on fossil evidence

Replacement hypothesis: suggest that archaic homo sapiens left Africa.

- Proposes a second migration happening about 100,000 years ago\

- Relies mostly on mtDNA evidence

MODULE 6: GENETIC CHANGE



Mutation

Inquiry Question: How does mutation introduce new alleles into a population?

- Explain how a range of mutagens operate, including but not limited to:
 - **Electromagnetic radiation sources**
Harmful radiation sources, called ionising radiation (X-ray, gamma ray), has enough energy to break chemical bonds. This leads to deletions, partial chromosome loss, rearrangements of sequences, interfering with cell division, gene production and cell metabolism.
 - **Chemicals**
Chemical mutagens are chemicals that cause mutations if cells are exposed to them at high frequencies or for prolonged periods of time. They affect the function of proteins, disturbing cellular processes.
Examples of chemical mutagens include
 - Ingested chemicals: alcohol, tar, nicotine, medications and chemicals in food, food additives and preservatives
 - Environmental irritants and poisons: organic solvents, cleaning products, asbestos, pesticides e.c.tThe effect of these mutagens usually results in the incorrect insertion of nucleotides.
 - **Naturally occurring mutagens**
Viruses: Mutations are caused when viruses insert their genetic information into the chromosomes of the host cell, causing disruption to normal cell function e.g. HPV is an **oncogenic (cancer causing) virus that causes cervical cancers**.
 - compare the causes, processes and effects of different types of mutation, including but not limited to
 - **point mutation**
Involves alterations to one (or more) bases in the DNA, such as bases being mistakenly inserted in place of another. Point mutations can also involve one or more bases being **inserted** or **deleted** from a DNA strand at a single point. Both insertion and deletion cause a massive impact as they result in a **frameshift**, where every codon from that point on is affected, which means that incorrect amino acids are being created, resulting in non-functional proteins.
e.g. **sickle cell anaemia is caused when DNA code is changed from CTC to CAC**.
What the DNA is changed to
 - **Silent:** no change to the final protein due to the redundancy of a codon-> amino acid table.
 - **Missense:** there is a change to the final protein (amino acid) e.g. sickle cell
 - **Nonsense:** a STOP codon is added early-> premature termination of sequence.
 - **Neutral:** the amino acid changes but to a similar property amino acid.
 - **chromosomal mutation**
Involves changes to entire chromosomes or significant parts of a chromosome. Chromosomal mutations can also include deletion of an entire chromosome, or sections of a chromosome-
Duplication, Inversion, Deletion, Insertion, Translocation.
e.g. **Down Syndrome is a chromosomal mutation when individuals have an additional copy of chromosome 21 in each of their cells**.
Deletion-> Part of chromosome is lost/deleted (reduced number of genes)
Insertion/duplication-> could be part of the same/different chromosome duplicating and adding on (increased number of genes-> redundancy) e.g. **Huntington's diseases**
Inversion-> part of chromosome removed, flipped and reinserted e.g. **Haemophilia**
Translocation-> moving parts of one chromosome to another non-homologous chromosome.
- Changes to chromosome number (Aneuploidy)**
- Cause by nondisjunction during meiosis
 - Non-disjunction is the failure of homologous chromosomes to separate during cell division (Meiosis 1- homologous chromosomes separating)
 - Outcomes are cells with extra or missing chromosomes

Trisomy: an extra copy of chromosomes is present. Can lead to shorted life-span or development problems.

e.g. **Trisomy 21 meaning three copies of chromosome 21, resulting in down syndrome**.

- distinguish between somatic mutations and germ-line mutations and their effect on an organism

Somatic mutation: Mutation to body cell

 - Some somatic mutations are compensated for by surrounding healthy cells. In other cases, the cells with the somatic mutation die and are replaced through the mitotic division of nearby healthy cells. However, some somatic mutations impact cell division and can cause uncontrolled cell replication. These mutations are associated with the growth of cancerous tumours.

Germ-line mutation: Mutation to gametes

 - If the mutation occurs in a gamete, then the mutation can be inherited by offspring. These germ-line mutations can give rise to new variations in a population.
- assess the significance of 'coding' and 'non-coding' DNA segments in the process of mutation

Coding DNA (Exons)-> sequence of DNA that codes for proteins and makes up 1.5% of the human genome. Mutations can affect this type of sequence, affecting protein production.

Non-coding DNA (Introns)-> Does not provide instructions for making proteins instead, it contains sequences that act as regulatory elements, determining when and where genes are turned on and off. A mutation in non-coding DNA can cause a protein to be expressed in the wrong time or can reduce or eliminate the expression of an important protein when it is needed.
- investigate the causes of genetic variation relating to the processes of fertilisation, meiosis and mutations

Genetic variation occurs due to:

 - *Random fertilisation:* random sperm fertilises a random ovum
 - *Random segregation:* homologous chromosomes separate randomly
 - *Independent assortment:* chromosomes separate into daughter cells independently (father and mother chromosomes are mixed into daughter cells)
 - *Crossing over:* new combinations of alleles so each gamete is unique

Mutations increase the types of alleles present in a population, also increasing genetic variation that can be detrimental (**causing death**), neutral (**no change**) or beneficial (**increase population**)
- evaluate the effect of mutation, gene flow and genetic drift on the gene pool of populations

Movement of alleles-> caused by migration and immigration (increases the genetic variation for all genes)

Increased allele frequency-> increases mutation-> increases genetic variation

Different effects on alleles

 - selective pressures
 - sexual selection
 - gene flow (immigration, migration)
 - genetic drift: Allele frequencies of a population change over generations due to chance, such as natural disaster. Those that remain survive by chance (*inaccurate, bottleneck*-> allele frequencies reduced, *founder*-> when a small group of individuals break off from a larger population to establish a colony)
 - SNPs (indicators of genetic variation in humans)

Biotechnology

Inquiry Question: How do genetic techniques affect Earth's biodiversity?

- investigate the uses and applications of biotechnology (past, present and future), including:

Ancient biotechnology

 - agricultural- selecting the best animals, cross breeding leading to changes in biodiversity.
 - Aquaculture- canal system and traps
 - Food production- bread and cheese making. Selective specific grains with fossil evidence. Yeast bacteria and fermentation processes introduced

Classical biotechnology

- Fermentation- yeasts ferments the sugars to produce alcohol and CO₂
- Medicine and antibiotic production- ancient remedies from plants creates awareness and leads to medicine being created from plants and fungi. Penicillium discovered by Alexander Fleming in AUS.
- Plant selective breeding- hybridisation and artificial pollination

Modern biotechnology

- DNA manipulation to precisely create end products
- Making specific proteins and cell processes to achieve specific outcomes
- Genetic engineering- involves changing DNA bases to change the end protein (e.g. recombinant DNA, GMO, transgenic species, DNA splicing involving restriction enzymes, DNA amplification using PCR, DNA recombinant using ligase enzyme)
- DNA sequencing- exact nucleotide sequence of DNA
- DNA profiling- amplification of short tandem repeats and gel electrophoresis

Future biotechnology

- Advances in genomics and how proteins are expressed in cells to improve treatment of diseases, cancer and genetic disorders.
- Producing environmentally friendly products
- Edible vaccines using micro-organisms, manufacture of synthetic cells, organs and transplants

– analysing the social implications and ethical uses of biotechnology, including plant and animal examples

Multiple aspects to consider:

- benefitting humans by making agriculture more productive
- breaching ethics if breeding 'superior genes'
- bioterrorism- misusing technologies
- tampering with nature- affecting nature evolution

When analysing social and ethical implications, things to consider:

- medical and health benefits
- financial and social justice issues
- animals and human rights
- effects on environment
- ownership
- religious values

e.g. Golden rice (check written notes)

– researching future directions of the use of biotechnology

CRISPR Cas9 enzyme that can precisely slice nucleotides for gene editing (check written notes)

– evaluating the potential benefits for society of research using genetic technologies evaluating the changes to the Earth's biodiversity due to genetic techniques

Biodiversity-> needed to maintain long-term healthy ecosystems for long-term survival and feeding human populations

- Creating GMO's and transgenic species can increase biodiversity by increasing the gene pool and gene combinations. In long-term, if this is continuously bred and reproduced, biodiversity will be reduced or wiped out.
- GMO's cross breeding with natural populations will affect natural biodiversity
- Can use in conservation techniques but genetically modifying organisms to make them disease resistant may be costly and time consuming
- Can compromise human populations with poor ecosystems/environment if gene variance is reduced, since genetic variation is needed for evolution.
- GMO's can outcompete natural species-> result in extinction of natural species

Genetic Technologies

Inquiry question: Does artificial manipulation of DNA have the potential to change populations forever?

- investigate the uses and advantages of current genetic technologies that induce genetic change compare the processes and outcomes of reproductive technologies, including but not limited to:
 - **artificial insemination:** collecting sperm from a chosen male and introducing into selected females by using artificial inseminations guns and inserting into vagina.
Issues: Still costly (buying equipment)
Benefits: can store sperm-useful for conservation
 - **selective breeding:** mating a male with a female with desirable characteristics in the hopes that the offspring will have some of the desired traits from both parents e.g. cows producing creamy milk bred with cows produced large quantities
Issues: can introduce undesirable traits (as its not precise), its time consuming and costly
 - **artificial pollination:** removing stamens of flowers and dusting pollen onto the stigma of a flower
Issues: Overuse of AP can lead to loss in biodiversity
Benefits: Creates new varieties and hybrid species increasing biodiversity, AP fills the gap left by the currently decreasing population of natural pollinators
 - same flower = self-pollination
 - different flower = cross-pollination
 - **IVF:** egg fertilised by sperm outside mothers body (in culture plate), zygote is left to proliferate/divide and then is transferred to mother (or surrogate)
Issues: useful to produce stud breeders (only producing eggs-> inserting zygotes into surrogates)
Benefits: genes for infertility can be passed on and genetic variation decreased
- investigate and assess the effectiveness of cloning, including but not limited to:
 - **whole organism cloning:**
 1. **Cutting and grafting- plants**
 - Cutting-> a section is removed from parent plant and placed in soil or water. It develops its own roots, stems and leaves, developing into a full-sized genetic copy of original plant
 - Grafting-> a cutting from the stem of the plant is bound to the cut stem of another plant with developed roots. The stems fuse and grow together
 2. **Tissue cultures- plants**
 - A section of the parent plant is pulverised releasing individual plant cells. Cells are grown on a nutrient and hormone containing medium. Cells grow to form small sprouts
 3. **Artificial embryo twinning- animal**
 - is fertilised by sperm, forming a cell. This cell develops into a clump of identical unspecialised cells which are split and form embryos. These embryos are implanted into new mother. The surrogate mothers give birth to genetically identical offspring.
 4. **Somatic cell nuclear transfer (SCNT)- animal**
 - A donor cell is taken from the organism to be cloned, and an unfertilised egg is taken from a female organism. DNA in the nucleus is removed from unfertilised egg in a process called enucleation and replaced with nucleus from donor cell. Egg cell is triggered to divide by an electrical impulse, developing into an embryo. Embryo is transplanted to surrogate mother who gives birth to a genetically identical copy of original organism.

Ethical issues:

 - I. Concerns for animal welfare
 - II. Religious argument “acting as god”
 - III. Expensive procedures
 - IV. Unforeseen health risks (controversial, since debates on Dolly the clones premature death
 - V. Moral and legal concerns if used on humans

Bigger issues:

 - I. Reduction in genetic diversity of populations
 - II. Only one parent-genetically identical offspring

- III. Cloned organism can become dominant, preserving the traits and letting other genes die out
- IV. Population less likely to survive sudden changes/disease

Application of Whole organism cloning- why do we clone whole organisms?

- *Scientific research*-> animal testing in the development of drugs and medical devices
- *Agriculture*-> cloning enables desirable traits to be passed onto offspring. Faster and more reliable than natural breeding.
- *Wildlife conservation*-> cloning can be used to prevent the extinction of critically endangered species. Cloning can also be used to bring back species from extinction

- gene cloning

- 1) **PCR**-> a technique used to make lots of copies of a specific region of DNA, in vitro (in a test tube) via thermal cycling

Steps involve:

1. **Denaturation:** DNA hydrogen bonds are broken separates to form single strands (95 degrees)
2. **Annealing:** Primers bind to DNA strands, forming sticky ends (55 degrees)
3. **Extension:** Nucleotides continue being added by forming complementary base pairs-> DNA polymerase synthesises new DNA (75 degrees)

- 2) **Recombinant DNA**-> DNA which contains genes from two or more different sources, in vivo (living organism) . Involves inserting desired gene into a bacterium. The bacteria divides and clones the desired gene.

Applications of gene cloning- why do we gene clone?

Direct applications

- DNA sequencing
- DNA profiling

Indirect applications: applications involving a cloned gene being inserted into an organism **i.e. transgenic organism**

- Gene analysis-> genes can be added or removed to research
- Gene therapy-> scientists replace a disease-causing gene with a normal gene (prevents or corrects expression)

➤ describe techniques and applications used in recombinant DNA technology, for example:

A) **Recombinant DNA**- Production

1. **Isolation:** DNA fragments are isolated from their natural sources
 - Target gene-> gene of interest **e.g. human insulin gene**
 - Scaffold gene-> DNA molecule that the targeted gene is going to be inserted into **e.g. plasmid (small circular piece of DNA found in bacteria. Plasmids can be naturally transferred between bacterial species)**
2. **Digestion:** DNA fragments are cut using the same restriction enzyme (**enzyme that cuts DNA bases specifically**)
 - Target gene and plasmid are cut with the same restriction enzyme to achieve sticky ends that are complementary to each other.
3. **Insertion:** Sticky ends bind so that the target gene is inserted into the scaffold DNA to form recombinant plasmid
4. **Ligation:** DNA ligase seals the backbone to produce a recombinant molecule
Recombinant plasmids have potential to cause issues, by not binding properly or correctly.

Positives

- Chance of finding suitable gene is improved
- Efficient in inserting desired trait
- Only affects desired trait
- Can be used in pest resistance
- Can be use in medical treatment

Negatives

- Accidentally species loss
- Decreased biodiversity
- Has potential to not bind correctly or properly

B) Bacterial transformation

To amplify the target gene, we introduce the recombinant plasmid into a bacterial cell e.g. *E. coli*

1. Combine plasmids with bacteria
 - Add both to a solution containing calcium ions
 - Calcium ions disrupt the bacterial cell membrane (holes) allowing recombinant DNA to move into cell
2. Heat shock the bacteria
 - Involves quickly increasing the temperature for a short amount of time
 - Forces recombinant DNA into cells through disrupted membrane
3. Allows the bacteria to recover
 - Bacteria are transformed into nutrient-rich broth at their optimal temp (~37degrees)

Selecting transformants-> Not many bacteria will be transformed despite the process

Transformant

- Only 1/1000 bacteria will take up a plasmid DNA

Non-transformant

- Won't take up plasmid by chance
- Bacteria are killed by shock

To find transformants scientists use plasmid with a selectable marker e.g. antibiotic resistance gene

▪ Other types of transformation of recombinant DNA:

1. *Microinjection*: DNA from one species is inserted into a cell from another species using a micro-pipette
2. *Biolistics*: a gene gun shoots small metal bullets coated in DNA into the nucleus of a cell
3. *Electroporation*: cells are quickly exposed to series of short electrical impulses, creating a hole in the cell membrane through which DNA can move into the cell
4. *Viruses*: by introducing a target gene into the viral genome, a virus can be used to insert a gene into an organism

Use of Transformants

1. "Protein factory"-> transformed bacteria as factories to make the protein which is encoded by the target gene. Protein can be harvested and purified e.g. insulin for diabetes
2. "Plasmid factory"-> used for other things e.g. DNA sequencing, DNA profiling, transgenic organisms

— the development of transgenic organisms in agricultural and medical applications

Transgenic organism -> a genetically modified organism that contains genes of another species

A transgenic organism is created by taking a gene from one species (a transgene) and inserting it into the DNA of another species.

Uses of transgenic organism

- To increase crop and livestock resistance: increased protection against pests, diseases and extreme environment conditions e.g. BT cotton
 - > contains a gene taken from soil bacterium
 - > genes codes for a toxin disrupting the digestive system of cotton-eating caterpillars
 - > Achieves higher yield for farmers without need to increased insecticides (saves money and time, lower environmental impact, lower impact on human health)
- To increases the productivity, yield and quality of crops and livestock e.g. golden rice
 - > Rice engineered with beta carotene genes from daffodil/maize and bacteria
 - > Beta carotene improves Vitamin A levels, combating Vitamin A deficiency related blindness
 - > Golden rice grown without proper maintenance may pass engineered genes onto other rice populations

- To produce pharmaceutical products e.g. insulin
 - > needed to treat people with diabetes
 - > human gene for insulin can be inserted into a plasmid of E. coli
 - > this insulin can be harvested and purified for medical purposes

- evaluate the benefits of using genetic technologies in agricultural, medical and industrial applications

Agriculture

 - creates possibility to produce crops and animal varieties that are better suited to specific environments e.g. high rainfall or drought
 - improving yield e.g. BT cotton
 - improving nutritional value e.g. golden rice
 - improving quality & quantity
 - improving resistance to disease

Medical

 - gene therapy & IVF helping with identifying genetic disorders e.g. choosing functioning, healthy embryos + delivering functional genes to people with mutated copies
 - Creating more precise restriction enzymes to improve other biotechnologies e.g. CRISPR CAS9

Industrial

 - pollution prevention- using engineered enzymes for breaking oils/wastes
 - biomaterial production- synthetic/natural substances for diagnostic/medical/therapeutic purposes
 - biofabrication- creating synthetic tissues/organs

- evaluate the effect on biodiversity of using biotechnology in agriculture

Biodiversity is needed to maintain long-term healthy ecosystems for long-term survival and feeding human populations

 - creating transgenic species can increase biodiversity by increasing gene pool and gene combinations HOWEVER in the long-term, if this is continuously bred and reproduced, biodiversity will be reduced- species can be wiped out
 - GMOs crossbreeding with natural populations will affect natural biodiversity
 - Can compromise human populations with poor ecosystems/environments if gene variance is reduced, since genetic variation is needed
 - GMOs can outcompete natural species-> result in extinction of natural species

- interpret a range of secondary sources to assess the influence of social, economic and cultural contexts on a range of biotechnologies

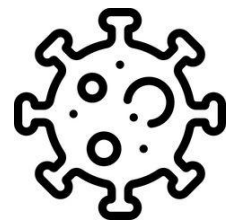
	Social
Whole organism cloning	Social implications of whole organism cloning involve how cloning technologies affect society. In 1997 when the first mammal, Dolly, the sheep was cloned, there was much social speculation on the efficacy of whole organism cloning. While many researchers and industries benefited from the mass production of crops and proteins with desired features, many animal-welfare activists protested against the nature of cloning. Animal-welfare activism is due to high failure rate, birth defects and surrogate mothers. Animal-welfare activist call to protest against cloning for research and agricultural uses, forming organisations such as the <i>Compassion in World Farming</i> and the <i>American Anti-vivisection Society</i> . Specifically, the <i>American Anti-vivisection Society</i> argue that 96-99% of cloning attempts regularly cause death, deformities or severe health problems. This devalues animal life by treating them as mere commodities. They primarily protest to “End Animal Cloning” seeking to keep meat and milk products from cloned animals and their offspring out of human and animal food supply. Their “No pet cloning” initiative seeks to protect both animals and people from being exploited by companies attempting to ‘bring back’ a deceased animal. Whole organism cloning is also being used for the revival of endangered species. For

	example, scientists cloned the endangered black-footed ferret, duplicated from the genes of an animal that died over 30 years ago. The cloned ferret, named Elizabeth Ann, is a genetic copy of ferret Willa who died in 1988 and whose remains were frozen in the early days of DNA technology. The successful reproduction of a cloned endangered species is a landmark in conservation genetic research, proving that cloning technology can not only help restore genetic diversity but also allow for future breeding, opening new possibilities for species recovery. The social response to cloning for conservationist purposes has generally been received positively. However, some raise ethical concerns about cloning being a manipulation of life, and its potential for creating “designer” animals. Cloning is also seen as a band aid solution rather than a real approach to biodiversity conservation. Another common argument against the cloning of endangered animals focuses on the sheer cost of such experiments. The process is expensive and time-consuming and current cloning techniques have an average success rate of less than 5%. Instead freezing animals DNA and cells would give an option to clone animals in the future as a conservation effort.
	Economic Whole organism cloning has massively impacted the global market and economy. Cloning of animals for commercial purposes include animal husbandry, medical research, competition animals, pet cloning and restoring populations of endangered and extinct animals. The Commercial Animal Cloning Market is currently valued at USD 512.2 million in 2023 and is projected to reach USD 948.8 million by 2031, growing at a compound annual growth rate of 8.6% during the forecast period of 2024-2031. The USA dominates this market with a share of 35% in 2022. However, there are opinions that the money and energy now being spent on the development of animal cloning is a terrible waste of financial and intellectual resources that could much better serve animal welfare interest if redirected to better causes. Whole organisms in agriculture has shown promising economic benefits, particularly in livestock farming. For example, studies indicate that cloning can increase milk production per cow significantly, leading to substantial revenue growth for dairy products. Additionally, cloned animals with superior genetic can enhance disease resistance and milk quantity, which contributes to sustainability and profitability. An example of this is the cloning of Holstein Friesian cows in China, where scientists successfully produced “super-cows” capable of yielding up to 18 tonnes of milk annually or 100 tones over their lifetimes. This breakthrough allows for the preservation of highly productive cows. This advancement can improve efficiency of milk production and reduce reliance and expense on importing cattle.
	Cultural The cultural implications of cloning are based around the social response towards the biotechnology, in regard to societal values, beliefs, and ethics. The ethical considerations are deeply influenced by religious beliefs, which vary widely across cultures. Centrally, they all argue that cloning challenges the natural order and raises questions about human dignity. For example, the Christian faith may oppose cloning, viewing it as a violation of divine will. In Judaism there is cautious openness to biotechnology provided it serves a beneficial purpose and adheres to ethical safeguards. Similarly, Islamic views generally allow for cloning if it addresses significant needs and avoid harm. Additionally, after the cloning of Dolly the sheep, a debate on the practical value and ethics of cloning became more popular especially on the discussion of possible human cloning. 81% of Americans say cloning a human being is not morally acceptable. Fewer Americans are concerned with cloning animals than with the prospect of cloning humans.

	Social
Recombinant DNA	Social implications of recombinant DNA show how the technology has impacted society. For example, researchers have experimented with recombinant DNA in cotton to improve its disease resistance. Currently, Bt cotton accounts for 93% of cotton grown in India. The effects of introducing Bt cotton in India have been debated, assessed and researched in the past 15 years. The broad consensus among scientists allude to a positive effect on cotton farmers' livelihoods and the environment as profits and yield has increased. However, the views about Bt cotton in India does not always reflect positive research results. Indian government have been reluctant to fully embrace the technology of genetic modification and farmers enthusiasm for Bt cotton has been questioned. There is also widespread controversy on the link between the introduction of Bt cotton to India, and a surge in farmer suicides. Similarly, recombinant DNA has socially impacted the field of medicine, through human insulin. Before recombinant insulin, diabetics relied on insulin from animal pancreas, which could be unreliable, lead to allergic reactions and was ethically questioned. Using animal insulin caused for debate on animal welfare, pointing out the cruelty of performing vivisection (practice of performing operations on live animals). The introduction of recombinant insulin has seen a discontinuation of sourcing animal-based insulin in majority of the world. Recombinant DNA technology ceased the debate against animal-welfare activists and medical professors of the efficacy of animals providing insulin treatment. Now with recombinant technology - in one study - proinsulin is produced in inclusion bodies at a 10% level in <i>E. coli</i>, yielding 1.3 mg/L of insulin. Bt cotton plays a significant role in improving farmers' incomes and reducing their workload, which can positively influence their overall quality of life. In China, the additional income earned from growing Bt cotton enables farming families to buy more food and enhance their diets, leading to better nutritional outcomes. In South Africa, the reduction in time spent on pest control, particularly by women, frees up time for more meaningful activities, such as caring for children or the sick
	Economic
	Economic implications of recombinant technology is shown through its impact on the global market and economy. The expenses surrounding growing normal cotton have reduced since the introduction of Bt cotton. Economically, Bt cotton has had a general positive impact as less crops are susceptible to disease, and healthier harvests are producing increased yield. Initially in India, Bt cotton has been estimated to have reduced chemical insecticide use by 22% and increased farmer profits by 68% for a short amount of time. In Australia Bt cotton has been commercially grown since 1996 and currently more than 99% of the cotton grown in AUS contains recombinant traits. Since the introduction of Bt cotton, there has been a 97% decrease in insecticide use. The Australian cotton industry, including Bt cotton, generates over \$3 billion per year in export earnings and provides employment for over 7,000 people. Industries have also benefit from the exportation of recombinant DNA species (transgenic species). The U.S. is now leading in Bt-cotton exports, and it has been observed that US farmers have received an additional economic gain of between US\$50 billion and US\$160 billion in the sixteen-year period after adopting genetic modification technology. India is the third largest Bt cotton exporter after the United States and Brazil. In 2023, the United States earned \$6.96 billion from cotton exports, while India earned \$6.34 billion, and Brazil earned \$3.2 billion. However, some research poses that Bt cotton has little effect on cotton yield, and although it may have led to an initial reduction in insecticide use, this effect is now diminishing due to insecticide resistance in insect populations. Additionally, purchasing seeds for Bt cotton have become another expense for farmers. Seeds for Bt cotton are majority (97%) sold by Monsanto, but Monsanto's seeds are becoming expensive and like other hybrids lose vigor after one generation, requiring farmers to buy new stocks every year. This creates a greater gap

	between the wealthy higher class and poorer working classes. Similarly, recombinant human insulin has affected global economy. The global human insulin market is expected to grow from an estimated USD 19.6 billion in 2024 to USD 40.8 billion in 2033, at a compound annual interest rate of 8.5%. The market expansion is due to the speed at which insulin came to be used to treat type 2 diabetes, which was previously thought as a non-insulin dependent diabetes. As the prevalence of type 2 diabetes increases, the number of insulin users have also increased. Many regions of the world, such as Asia, are currently experiencing a disproportionately large increase in the prevalence of diabetes, both type 1 diabetes and type 2 diabetes, due to more effective screening, enhanced early diagnosis, and changes in lifestyle due to rapid economic growth and urban development. Recombinant human insulin has an estimated production cost of US \$2.28-3.37 per 1000-unit vial. According to the standard 40 units per day of insulin for a diabetic, a year's supply of insulin should cost less than US\$71. Though regular recombinant human insulin is affordable, a specific type of recombinant insulin, analogue insulin, has become unaffordable. Analogue insulin is based off the structure of human insulin, but have some different properties designed for slower or faster absorption or prolonged duration of action. Due to insulin often having to be purchased privately in most parts of the world, people in the world's richest countries struggle to pay for their insulin, whereas those in poorer countries may not be able to obtain it at all.
	Cultural Cultural implications are seen through recombinant DNA technologies impact on the cultural values and ethics of society. Recombinant DNA technology has significantly influenced cultural beliefs and ethical considerations, particularly in healthcare and agriculture. For example, the production of synthetic insulin through recombinant DNA has transformed diabetes management. Before this innovation, diabetic patients relied on animal insulin, which caused allergic reactions in about 5% of cases. It was also not equitable to patients who had cultural and religious guidelines which prevented the use of animal products. Animal insulin was also not received well by animal enthusiasts, vegan/vegetarians and other individuals. Recombinant DNA technology now enables the production of human insulin, reducing complications and improving the quality of life for over 537 million diabetics globally. In agriculture, recombinant DNA crops like Bt cotton have increased yields by up to 30% in India, benefiting farmers economically. However, these advancements have also raised ethical concerns, such as the exploitation of seeds by corporations like Monsanto, which controls over 90% of the recombinant seed market in some regions, creating economic disparity.

MODULE 7: INFECTIOUS DISEASE



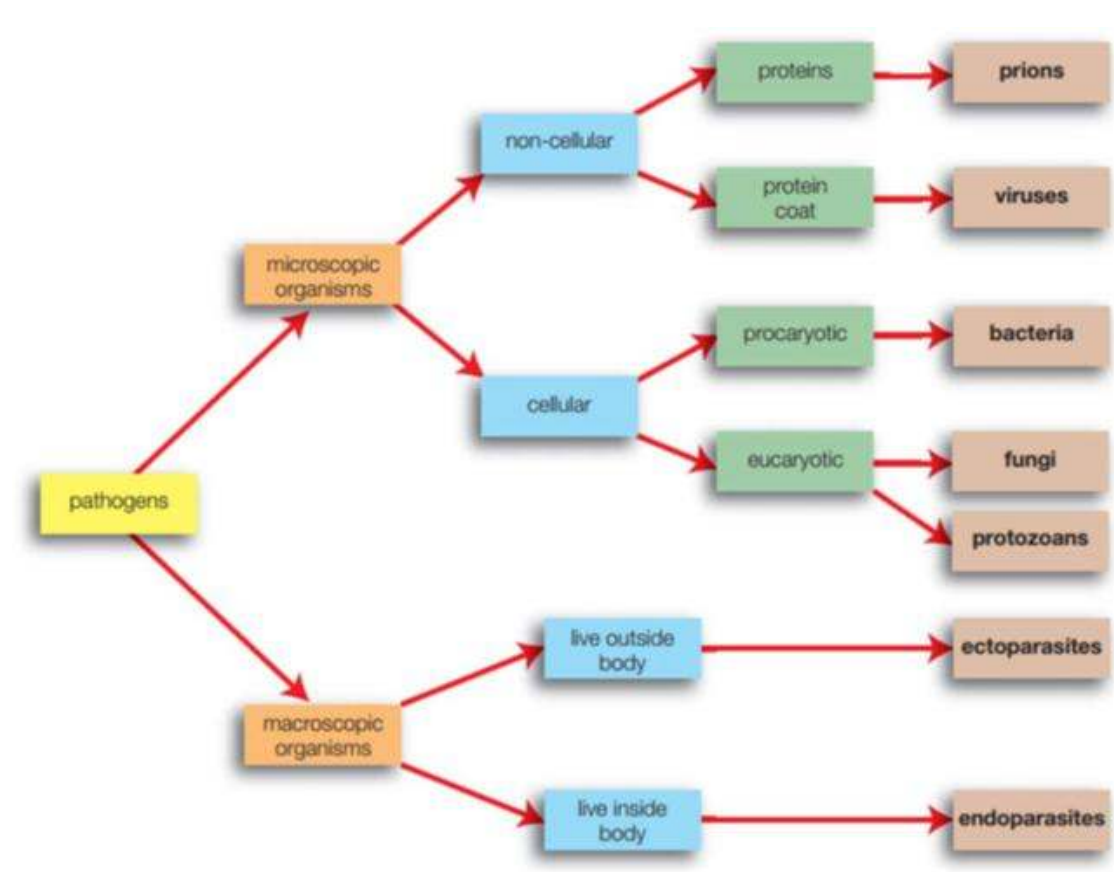
Causes of Infectious Disease

Inquiry question: How are diseases transmitted?

- describe a variety of infectious diseases caused by pathogens, including microorganisms, microorganisms and non-cellular pathogens, and collect primary and secondary-sourced data and information relating to disease transmission, including:

Infectious disease-> disease which can be transmitted from one organism to another, and are caused by pathogens

- **Pathogens:** any organism that can cause disease



- classifying different pathogens that cause disease in plants and animals

Pathogen	Key disease mechanism	Transmission	Examples
Bacteria-> prokaryotic, unicellular organisms cellular	They reproduce via binary fission. Their genetic material is in two forms: as bacterial DNA in circular chromosomes, and as smaller circular fragments called plasmids. Pathogenic bacteria can cause disease usually releasing toxins or damaging host tissue.		Tuberculosis (caused by Mycobacterium tuberculosis. Enters lungs when a person inhales infectious droplets) Tetanus Grown Gall (plants)
Fungi-> Eukaryotic, non-photosynthetic (with cell wall) cellular	Fungi are constructed by filaments called hyphae, and create structures called mycelium. Damage caused by secreting digestive enzymes and chemicals into their surrounding matter.	Cutaneous, subcutaneous, systemic (most lethal). Direct contact transmission or through contaminated objects	Athletes foot (caused by microsporum, Epidermophyton, and trichophyton. They live in outside layer of human skin. Symptoms-> itchiness, flaky skin)
Protozoa-> Eukaryotic, unicellular microorganisms (no cell walls) cellular	They are a sub-group of unicellular Protista that reproduce via binary fission.	Via vectors or contaminated water	Malaria (caused by plasmodium which feeds on haemoglobin, causing them to pop. Symptoms> fever, muscle pain, nausea e.c.t) Phytophthora dieback (plants)
Viruses-> non-cellular entities consisting of nucleic acid (DNA/RNA) encased in a protein coat (capsid) Non-cellular	Needs host to replicate. Virus attaches to host cell which is then penetrates to make its own viral proteins and nucleic acid, making copies of itself to infect other cells.		Influenza (caused by the Influenza virus which damages human lung tissue. Most symptoms are caused by the bodies own immune response including fever, coughing, and nasal consumption.

Prions -> Non-cellular infectious proteins which are abnormally folded versions of a protein. Non-cellular	Prions can lead to serious problems depending on structure, function and location of normal protein. Usually affect brain and neural tissue, which leads to neurodegeneration		Kuru (caused by a prion found in contaminated human brain tissue. Transmitted by eating brains of infected persons. Symptoms-> shivering, trembling, uncontrollable laughter)
Macroscopic parasites -> multicellular pathogens which can be seen with the naked eye cellular	They tend to cause disease by competing for nutrients from the host, also producing toxins and damage host tissue		Ticks Fleas Tapeworm

– investigating the transmission of a disease during an epidemic

Endemic-> a disease is transmitted but the number of cases remain unchanged

Epidemic-> a disease is transmitted but the number of cases increases e.g. pink eye (conjunctivitis)

Pandemic-> when the epidemic occurs in several continents, it is a global epidemic e.g. COVID (originated from China), Black Plague

– investigate modes of transmission of infectious diseases, including direct contact, indirect contact and vector transmission

Transmission involves carrying of transferring of a pathogen from an infected host to a non-infected organism through a “chain of infection” involving:

- Host that is susceptible to disease
- Pathogen capable of causing disease

Modes of transmission

Direct contact-> physical contact between infected organism/host and susceptible organism/non-infected organism

- touching
- sexual contact
- kissing
- contact with nasal/oral secretions
- direct contact with wounds

Horizontal transmission: contact between organism of the same generation, or between organisms *that are not parent and child*.

Vertical transmission: contact between offspring and parent

- *Prenatal* (before birth or during pregnancy): transplacental-> to a foetus from a mother through placenta
- *Perinatal* (around time of birth): during vaginal birth-> the microflora present in a mothers cervix and vagina is passed onto the child as it passes through the birth canal, such as an STI.
- *Post natal*: breast feeding, fluids and nutrients are passed to the baby through milk.

Examples-> Ebola transmitted through infected blood, glandular fever, ringworm, HIV/AIDS, herpes

Indirect contact-> Occurs when the host and another organism have no direct contact with each other.

- Airborne transmission

- Touching surfaces
- Contaminated food/water
- Infected surgical instruments
- Vectors (mosquitoes and ticks) causes disease such as malaria.

Examples-> Black spots on plants caused by fungus spores blowing from the wind, measles and influenza cause by infected droplets/bodily fluids, gastroenteritis cause by E. coli in contaminated food/water.

- investigate the work of Robert Koch and Louis Pasteur, to explain the causes and transmission of infectious diseases, including:

– **Koch's postulates**

Koch developed the agar plate technique for growing micro-organisms and used it to culture isolated anthrax bacillus.

Anthrax bacillus-> He examined the blood of a sheep that had died from anthrax and identified rod-shaped bacteria which he isolated and grew in cultures. He injected cultured bacteria into healthy sheep which subsequently developed anthrax. This supported germ theory.

Kochs postulates

- 1) The same micro-organisms must be present in every disease host
- 2) The micro-organisms must be isolated and cultures in the laboratory and accurately described and recorded
- 3) When a sample of the pure culture is inoculated into a healthy host, this host must develop the same symptoms as the original host
- 4) The micro-organism must be able to be isolated from the second host and cultured and identified as the same original species.

– **Pasteur's experiments on microbial contamination**

Microbial fermentation-> Pasteur discovered that microbes such as yeast and lactic acid bacteria are responsible for fermentation/

Pasteurisation-> Technique used to kill the germs he had discovered by boiling wine/beer/milk (essentially sterilising) before cooling

Proof of germ theory

- 1) Pasteur prepper a bone broth and placed equal amounts into two glass flasks. One had a straight neck; one was drawn out (creating a swan neck appearance)
- 2) He sterilised it by boiling it
- 3) Liquid in the straight-necked flask become contaminated, meanwhile the swan neck flask had no contamination

Germs in the air reached open flask, not swan neck supporting germ theory and eradicating the spontaneous generation theory.

Vaccination-> Pasteur discovered how to make vaccines by administering weaken versions of the pathogen to susceptible individuals. He developed vaccine for chicken cholera, anthrax and rabies.

- assess the causes and effects of diseases on agricultural production, including but not limited to:

Factors contributing to risk of infectious diseases include:

- increased mobility of humans (travellers, imports)
- rise of intensive and industrial type agriculture
- changing patterns of land use (forestation/irrigation changing)
- climate change-> changing abundance of insect vectors (due to changes such as rainfall, temp)
- antimicrobial/pesticidal resistance due to overuse

- loss of genetic diversity->monoculture/inbreeding leading to loss in resilience
- **plant diseases**

Plant disease- causes:

- pathogens attack plants, entering through stomata, wounds, roots
- more common in higher densities of plants where soil, pH and water is NOT optimal (plants are stressed)

Fungi:

- common cause, reservoirs in machinery, soil, seeds and weeds
- transmitted by water, wind, contact with reservoirs
- damage the plants by destroying conducting vessels and absorbing nutrients
- e.g. rust and mildew

Insects and Mites

- acting as vectors or cause direct damage

Bacteria

- reservoirs in soils, weeds, humans
- multiply and spread in humid, warm weather, overcrowding of plants and inappropriate soil conditions (pH, water, nutrients)

Viruses

- less understood e.g. tobacco mosaic virus
- form reservoirs, stable in environment and persist in plant material after cropping

Nematodes (worms)

- Only some act as pathogens, causing wilting discoloration and death
- Require repeated cultivation and exposure to sun to remove reservoirs

Phytoplasmas

- Related to bacteria, but no cell wall
- Transmitted by vectors, inhibit phloem tissue

Plant disease- Effect:

- Biological (wilting, discoloration, death)
- Social and economic effects of AUS economy
- Social and economic effects of farmer(reduced yield loss, loss of trading opportunities, financial hardship and stress)

- **animal diseases**

epizootic-> animal equivalent of epidemic

zoonosis-> transmission from animal to humans

Animal disease-effect

- death of infected animal
- loss of appetite and weight
- economic loss
- zoonosis

Examples of disease

Bacteria-> foot rot, tuberculosis (mycobacterium tuberculosis, mycobacterium bovis)

Virus-> hand, foot and mouth, coronavirus, avian influenzas

Fungi-> ringworm, aspergillosis

Protozoa-> toxoplasmosis

Prions-> bovine spongiform encephalopathy

Microparasites-> roundworm, fluke

- compare the adaptations of different pathogens that facilitate their entry into and transmission between hosts

Adaptations to facilitate entry into hosts

Bacteria	Adhesion: – Pilli and fimbria: these adhesions allow for the surface of the bacterial cell to resist the secretions of urine, mucus and cilia that wash out pathogens – Bacterial cell forms a biofilm Invasion: – Enzymes break down cell contents – Resist phagocytosis by host cells – Toxins are secreted to damage host cell (endotoxins and exotoxins)
Fungi	Adhesion: – Assisted by cell wall or capsule molecules that permit adhesion to host cells Invasion: – Thermotolerance – Cell wall and capsules protect fungi from host attacks e.g. alpha glucan is a cell wall polysaccharide that gives protection – Secretion of hydrolytic enzymes cause damage to host cells and provides nutrients for fungus – Opportunistic fungal infections are common in immunosuppressed patients e.g. HIV/AIDS, chemotherapy
Virus	Adhesion: – Must enter the nucleus of the host cell to facilitate replication of the viral genome – Viral surface proteins adhere to host cell surface receptors and co-receptors Invasion: – Receptor-mediated endocytosis (movement of the virus into the cell). Enveloped viruses e.g. influenzas, are enclosed within an envelope (endosome) formed from the host cell membrane as they move into the cell. – Non-enveloped viruses e.g. polio virus, form a pore in the host cell membrane and deliver the viral genome through it
Protozoan	Invasion: – Form molecules to enter into a cell and attach to receptors – Use motile force to drill into host cell\ e.g. toxoplasmosis: microtubule protrusion into host cell facilitates entry (intracellular parasite) and formation of vacuolar membrane gives protection from lysosomes. Changas disease: receptor-mediated attachment, recruits lysosomes to fuse with cell membrane. Pathogen enters vacuole made of lysosomal membrane, then deactivates lysosomal enzymes.
Prions	Invasion:

	– hosts immune system secrete factors that enables prions to enter lymph tissue and spread to nervous system
Macroparasites	Invasion: – attached by specialised mouthparts – various hooks and suckers for attachment to or inside the body

Adaptations to facilitate transmission between hosts

Airborne	– able to remain suspended in air- lightweight – resist drying out- aerotolerant – pathogen causes sneezing and coughing, which causes ejection and transmission to new host	e.g. influenza
Waterborne	– able to colonise and proliferate in water, so environmental reservoir is present – outer surface structures to allow mobility e.g. flagella – cannot be affected by boiling	e.g. Giardia, Vibrio
Vector-borne	– Vector is not affected by pathogen – Form reservoirs in digestive tract of salivary gland of vector for easier transmission	e.g. malaria
Faeco-oral	– Pathogens are generally very stable in varied environments e.g. acid in stomach, low oxygen of large intestine – Induction of vomiting and diarrhoea increases likelihood of transmission	e.g. E.coli, salmonella
Soil-borne	– Form endospores to resist desiccation – Stable on the environment under a range of conditions	e.g. fungi, nematodes

Responses to pathogens

Inquiry question: How does a plant or animal respond to infection?

- Investigate the response of a named Australian plant to a named pathogen through practical and/or secondary-sourced investigation, for example:

Plant Defences

Passive defences: General strategy for preventing entry

- *Physical barriers:* thick cuticle, cell wall, small stomata, bark inhibit pathogen entry. Vertical hanging leaves reduce reservoir build up
- *Chemical barriers:* chemical components such as saponins, reduce pathogen growth and ward off vectors. Plants produce enzymes that break down pathogen derived toxins. Chemical receptors detect the presence of pathogen via PAMPS (pathogen-associated molecular patterns)

Active defences: more targeted approaches when pathogen infect

- *Pathogen recognition:* detecting physical and chemical signals. Regulated by genes
- *Rapid Active Response:* recognition of pathogen changes permeability in cell membrane-> movement of ions trigger defense responses e.g. releasing hydrogen peroxide or reinforcing cell wall or apoptosis.
- *Delayed Active Response:* involves limited spread of pathogen e.g. repairing wounds by producing cells and gum secretion. Salicylic Acid can help with signalling memory-> systemic acquired resistance.

- fungal pathogen

Myrtle Rust

Pathogen-> the fungus *Austropuccinia psidii*

Affected plants in the myrtaceae family

- eucalyptus
- bottlebrush
- tea trees
- lillypilly
- paper backs

Description

- Fungal infection
- Infection begins with small purple spots on leaves resulting in bright yellow spores which are dispersed by the wind
- Results in high rate of transmission
- Was first detected in 2010 and has since spread across eastern Australia
- fungal pathogen

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Host response 1

- Mechanical barriers
 - Bark
 - Thick cell walls made of pectic and lignin
 - Waxy leaf cuticles
- At site of infection, cell walls become thicker and stronger i.e. reinforced by additional structural proteins- this seals off the site of infection preventing further spread
- Secretory cells and glands transport defensive substances to the site of infection producing unfavourable conditions for the survival of Myrtle Rust fungus

Host response 2

- Pathogen is recognised as 'non-self'
- Innate and non-specific immune response begins
- Harmful metabolites accumulate in site of infection
- Fast release of hydrogen peroxide in an oxidative burst
- Reinforcement of cell wall
- Hypersensitive response (HR) deliberate plant cell death as site of infection (apoptosis)

– viral pathogens

- analyse responses to the presence of pathogens by assessing the physical and chemical changes that occur in the host animal's cells and tissues

Antigens- a substance that induces the formation of antibodies, therefore stimulating an immune response. Antigens are proteins on the surface of pathogens and are how the immune recognises threats. Often invading pathogens are referred to as antigens. The immune system doesn't affect self-cells because our cells contain self-antigen markers, indicating that they are not foreign particles.

Antibodies- proteins which are produced in response to the presence of an antigen in the body. Are Y shaped with two binding sites.

Lysosome- contain digestive enzymes that break down engulfed pathogens and cellular waste, facilitating immune responses.

Phagocyte- antibodies develop structure that recognise and bind to specific antigens

Lymph- Fluid that leaks out of capillaries into spaces between cells; circulates in the vessels of the lymphatic system

Lymph node- small structures located on lymphatic vessels where pathogens are filtered from lymph and destroyed by lymphocytes

Lymphatic system- system of the body consisting of organs, vessels, nodes and lymph that produces lymphocytes (in the bone marrow) and filters pathogens from body fluids

PAMPs- pathogen-associated molecular patterns, that are recognised by the innate immune system.

FIRST LINE OF DEFENCE

Physical barriers->

Skin: epithelial tissue consisting of three layers: outer epidermis, dermis and the lower hypodermis	Keratin-> found in epidermis. Is a waterproof layer that strengthens the skin against breakage and infection. Epidermis-> consists of flattened and dead skin cells that are tough and closely packed together forming an effective barrier against pathogens.
Mucus membranes: Line respiratory, urinary, digestive and reproductive tracts	Trap unwanted pathogens and substances until they can be removed by cilia. Contain lysozyme to help break down microbes
Mucus: Lining internal cavities	Secrete substances to kill pathogens. Form a barrier to prevent pathogen entry.
Cilia: small, hair-like projections.	Extend from the epithelial cells which line the respiratory tract. They sweep the mucus along the tract so they can be coughed, sneezed out or swallowed.

Chemical barriers->

Lysozyme:	enzyme which breaks down the cell walls of bacteria, killing them to fight infection e.g. tears, saliva, mucus
Acidic secretions:	Body secretions also chemically screen out pathogens by being very acidic e.g. vagina, ear canal, stomach, skin
Urine:	Urine is sterile until it reaches the lower urethra. Flushing action of urination assists in keeping pathogen away from bladder. Urine has some antimicrobial peptides and acidic pH. When not urinating, there is a chance for microbes to move up. Females have higher risk for infection due to a smaller urethra. A type of infection is bacterial cystitis.
Sebum and sweat:	Waterproof and lubricate skin. pH of sweat slightly acidic due to lactic acid amino acids and fatty acids. There are also lysozymes in sweat
Saliva: produced in salivary glands	Water, mucus, enzymes and antimicrobial substances that can flush and break down microbes

Tears: produced by lacrimal glands	Antimicrobial peptides, immunoglobulins, lysosomes and mucus.
Gastric/stomach secretions	Secretions of HCl creates a highly acidic environment discouraging growth and survival of pathogens. Pepsin in the secretions is antimicrobial.

Physiological barriers->

Microflora: found in and on mouth, skin and intestines.	They aid body by: ○ Competing for space and nutrients-> the microflora use up the space and nutrients causing pathogens to be unable to establish itself. ○ Producing antimicrobial chemicals-> prevent growth of other organisms e.g. the acidity of the skin results forms the activity of microflora, which break down the sebum in skin to form fatty acids.
Sphincter: circular muscle that maintains constriction of the natural body passage by contracting and relaxing muscles to separate different compartments.	Sphincters help to physically seal off compartments in the body to reduce the likelihood of pathogen invasion.
Peristalsis: contraction of muscles in a coordinated way to move food in one direction only.	Stasis (lack of movement) in the intestines can lead to intestinal bacterial overgrowth.

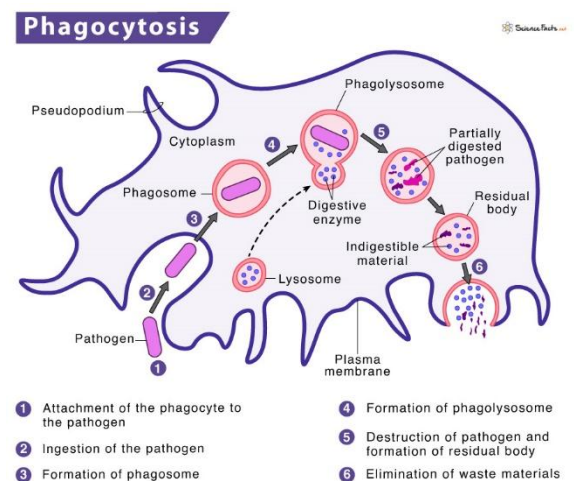
SECOND LINE OF DEFENCE

Inflammation-> accumulation of fluid, plasma proteins and white blood cells that occurs when tissue is damaged or infected.

The inflammation response is a non-specific defence mechanism and occurs at the site of infection. When the cells are exposed to a pathogen or are damaged (release damage-associated molecular patterns [DAMPs]), they release chemical signals; histamines (trigger vasodilation and vascular permeability), serotonin and prostaglandins (pain and fever of inflammation). These chemicals cause capillaries to dilate increasing blood flow to the site of infection or injury and causing the area to become show signs of inflammation: *pain, heat, redness, swelling, loss of function*. The increase in permeability of the blood vessels allow certain WBC's (neutrophils, macrophages, dendritic cells) to move into tissues and attack the pathogen.

Phagocytosis-> The process by which phagocytes engulf and destroy foreign or unwanted material

1. Phagocyte engulfs the foreign material, forming a phagosome.
2. A lysosome fuses with the phagosome, forming a phagolysosome
3. Within the phagolysosome the enzymes break down the foreign material into smaller pieces.
4. This residual vesicle fuses with the plasma membrane and the contents are released from the phagocyte by exocytosis.



- **Neutrophils**-> most abundant WBC. They quickly enter tissues and phagocytose pathogens in acute infection. They also expel antimicrobial compounds such as hydrogen peroxide which disrupts bacterial and fungal cell membranes, killing them. They also release cytokines, which attract other immune cells and promotes inflammation.
- **Macrophages**-> live longer and fight chronic infections. They are antigen presenting cells, and also release cytokines
- **Dendritic cells**-> play a role in phagocytosis and are antigen presenting.
- **Eosinophils**-> smaller in abundance, involved in allergic reactions and parasite disease. Cytotoxic attaching to pathogens, releasing substances to kill
- **Natural killer cells**-> patrol the body, and are important in defence against virus-infected and cancerous cells
 - They release cytotoxic chemicals when they are in close proximity to the target cell e.g. the chemical perforin punches holes in the cell membrane of the target cell, killing it.

Fever (Pyrexia)-> Our normal body temp is 37 degrees. The body reacts to pathogens by releasing pyrogens that cause fever. Purpose of fever is to lull or limit growth of pathogen and products. WBC's also work

Complement system-> group of proteins assisting defence in destroying pathogens. They can stimulate phagocytes to become more active, attract phagocytes to site of infection or destroy membranes of invading pathogens.

Cytokines-> they are chemical messengers that are produced during infection and promote the development and differentiation of T and B lymphocytes for the third line of defence. They also help stop replication of cells, promotes apoptosis and reduced protein synthesis. Cytokines play a role in the feelings of lethargy, muscle pain and nausea when you experience an infection.

Immunity

Inquiry question: How does the human immune system respond to exposure to a pathogen?

- Investigate and model the innate and adaptive immune system in the human body

B lymphocyte- a type of WBC which is produced in the bone marrow and accumulate in the lymphoid tissue.

T lymphocyte- a type of WBC which is produced in the bone marrow and accumulate in the thymus gland.

THIRD LINE OF DEFENCE

Antibody-mediated response/humoral response

When antigens are present, helper T-cells bind to the antigen and release substances such as cytokines which activates B cells. This causes the B cells to proliferate and divide into:

- **Plasma cells**- produce antibodies that bind to specific antigens forming an antibody-antigen complex.
 - ➔ **Antibodies:** interfere with the functioning of a pathogen, making it unable to cause damage, or make it easier for other components of the immune system to destroy it.

Neutralisation	Precipitation	Agglutination	Complement activation	Opsonisation
Deactivating pathogen or toxin by blocking its active site	Precipitation of dissolved antigens	Antibodies bind to antigens on the surface of cells to form clumps of cells	Activation of the complement system, leading to lysis of infected cell	Enhanced phagocytosis by natural killer cells

- **Memory cells**- provide immunological memory, providing body with long term defence against antigens. If the animal is exposed to the same antigen again, memory cells

recognise it and divide to produce antibody-producing plasma cells. This aids in efficiency of the immune response.

Cell-mediated response

Special types of T lymphocytes target and destroy the entire infected host cell, along with the pathogens inside them. T lymphocytes make direct contact with infected cells via special receptors and can also identify tumour cells and destroy them. There are several types of T lymphocytes:

- **Cytotoxic (Killer T cells)**- kill foreign infected and abnormal cells. They produce many copies of themselves and move to the site of infection. They then secrete or inject toxic chemicals (cytokines) into the target cell.
- **Helper T cells**- promote the activities of other immune responses by secreting cytokines which activate both the humoral response and cell-mediated response, increasing the activity of phagocytes, promoting inflammations and stimulating B lymphocytes to differentiate into plasma and memory cells.
- **Suppressor cells**- responsible for turning off the immune response after the antigen has been successfully contained, destroyed or removed.
- **Memory T cells**- provide body with long term defence in the body by cloning T cells that remain in the body to respond to future exposure from pathogens.

- Explain how the immune system responds after primary exposure to a pathogen, including innate and acquired immunity

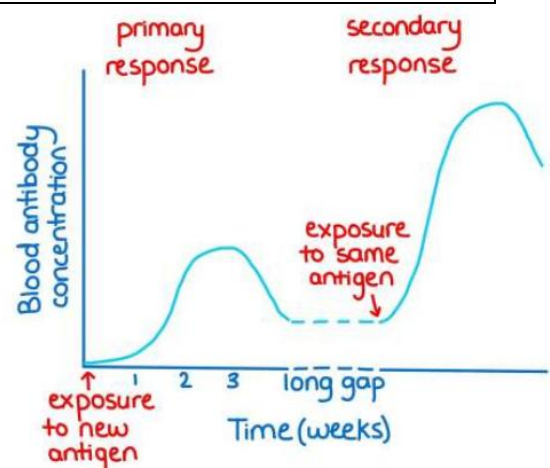
Immunity-> when the body is able to protect itself without symptoms arising

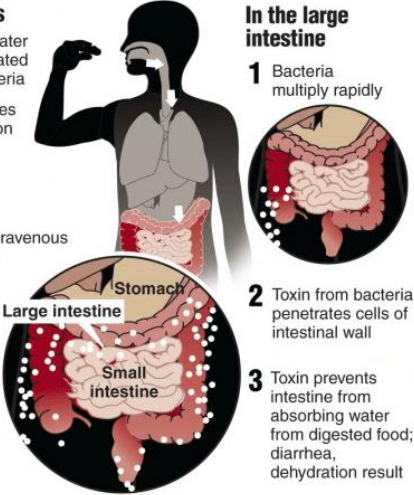
Secondary exposure

- Innate: responds the same way as primary exposure
- Adaptive: Memory B or T cells detect presence of a memorised antigen. Memory B cells alert helper T cells about re-infection.
 1. Memory B cells differentiate into plasma B cells which produce large quantities of antibodies fast so that signs of disease never develop (humoral response)
OR
 2. Helper T cells are activated, differentiating into cytotoxins to kill off pathogens (cell mediated response)

Case Study: The immune response after primary exposure

Pathogen	Disease	Immune response
Bacteria	<i>Vibrio cholera</i> (cholera)	Innate - production of cytokines, migration of neutrophils into the gut lining and the production of antibacterial peptides (e.g. defensins) that kill bacteria directly or possibly alter host gene expression to induce more cytokines to induce inflammation



		Adaptive- IgA (immunoglobulin/antibody) is secreted by the gut lining to protect it from bacterial colonisation. After a week there is an increase in B cells specific to <i>V. cholera</i> antigens. Serum antibody peaks at 1-3 weeks after initial infection and decreases after about a year. Memory B cells form for secondary exposure. How it spreads • People ingest water or food contaminated with cholera bacteria • In epidemic, feces of diseased person is source of contamination Treatment • Salt solution, intravenous fluids, antibiotics • In unprepared communities, death rates can be as high as 50 percent In the large intestine 1 Bacteria multiply rapidly 2 Toxin from bacteria penetrates cells of intestinal wall 3 Toxin prevents intestine from absorbing water from digested food; diarrhea, dehydration result  © 2010 MCT Source: World Health Organization
Fungi	<i>Trichophyton rubrum</i> (tinea)	In host tissues affected by tinea, enzymes (keratinases) break down skin barrier, inhibit the immune response and reduce replacement of superficial skin cells. Keratinocytes express TLR's (toll-like receptors) and defensins as a first-line response. Dendritic cells suppress fungal growth and stimulate cytotoxic T cell production. Cytokines are released and phagocytosis of fungal cells with proliferation of cytotoxic cells.
Virus	Dengue Fever	Virus infects macrophages and dendritic cells. After evading the innate immune system, the virus becomes blood borne and spreads around the body. Dendritic cells recognise viral PAMPs and release interferons. Interferon activate the adaptive immune system which leads to both T and B cell response.
Protozoa	<i>Plasmodium falciparum</i> (Malaria)	Innate- some people are inherently protected from the parasite due to a change in their haemoglobin or RBC e.g. people with sickle cell anaemia. Natural killer cells increase as they cause lysis (death) of parasite-infected RBC's. Killer cells also produce interferon leading to macrophage activation. They also produce chemokines which activate phagocytosis. Adaptive- Antibodies (IgM and IgG) are produced by plasma cells. Helper T cells and cytotoxic T cells play a role in parasite control. Repeated infection leads to different stages of immunity - antiparasitic immunity-> prevents pathogen getting in blood - Anti-disease immunity-> no clinical signs observed despite infection

		– <i>Premonition</i>-> protection from new infections due to a low-grade infection of the blood with malaria parasites. Opsonisation leads to increased susceptibility to phagocytosis, destruction by cytotoxic T cells or parasite inhibition by neutrophil and macrophages.
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Prevention, Treatment and Control

Inquiry question: How can the spread of infectious diseases be controlled?

- investigate and analyse the wide range of interrelated factors involved in limiting local, regional and global spread of a named infectious disease

Influenza A

Local spread-> environments such as schools, hospitals, or within one or few suburbs. Depending on the conditions of those environments, the spread of disease will vary.

Sections of a local area (e.g. suburbs) with high population density will have higher incidence of Influenza A than sections of the same local area (e.g. suburbs) with low population density or interaction.

Regional spread-> Spread can occur through the use of aircraft as it transfers affected and unaffected individuals to different regions or countries as people travel for work, holiday. Individuals may also live in areas that are more isolated, which would mean that they will come in contact with less infected individuals compared to those living in the city.

Global spread-> Aircraft are vessels that transport affected and unaffected individuals to different regions. Therefore, diseases such as influenza A can spread from one country to another at a massive scale. Disparity in the access to valuable information about methods that can be used to prevent or control the spread of disease may vary across countries. Some countries may have limited access to information, e.g. information in third-world countries are considerably limited compared to developed countries.

Common factors that apply on Local, regional and global spreads of Influenza->

- the availability of healthcare facilities e.g. hospitals, medical centres, access to vaccination
- increased use of antivirals have rendered to vaccines against Influenza A ineffective, such as amantadine, and rimantadine which were introduced as antiviral drugs against Influenza. However, they are not longer produced as the virus is now resistant to those antivirals drugs due to overuse therefore, no longer affective against Influenza A.

- investigate procedures that can be employed to prevent the spread of disease, including but not limited to:

- **hygiene practices**

Personal hygiene: reduced risk of pathogens entering our bodies, or transmission of these pathogens to others. Inhibits the build-up of micro-organisms on our bodies. Practices of personal hygiene include:

- Hand washing-> prevent spread of pathogens that cause symptoms such as diarrhoea
- Washing of body and hair-> prevent build-up of bacteria
- Brush teeth-> build-up of bacteria in mouth can cause gingivitis (swollen, bleeding gum)
- Sneezing into tissue/elbow-> prevents airborne droplets from spreading to others.

Community hygiene: prevents the build-up of pathogenic organisms in the community.

Practices of community hygiene include:

- Sewage and garbage disposal-> reduce the risk of pathogen numbers increasing and spreading throughout the community. Sewage may incorporate UV disinfection for management of pathogens in wastewater
- Sterilisation and disinfection of equipment-> hospitals, doctors surgeries, dentists, and hairdresser/barbers reduce risk of the spread of pathogens from one person to another
- City planning-> reduce overcrowding, reducing risk of transmission- important in controlling disease

Cleanliness of food and water-> Food utensils must be clean and disinfected. Foods should be cooked to a healthy amount, e.g. [Salmonellosis caused by salmonella bacteria is transmitted in undercooked food](#). The spread of contaminated water hosting pathogens results in many infectious diseases e.g. [Cholera is transmitted via water-borne bacteria called Vibrio cholerae](#).

– Quarantine

The role of quarantine is to minimise the risk of exotic pests and diseases entering AUS in order to protect our native flora and fauna, our agricultural industries, our environment and our health.

Animal quarantine-> involves all animals that come into AUS spending time at quarantine stations to ensure that they are free of disease.

Plant quarantine-> involved examining all plants, and plant products (fruit, seeds, cuttings, bulbs, wood) brought into AUS. Many are refused entry, however in some cases plants/plant items are only allowed after treatment by quarantine stations.

Human quarantine-> the captains of aircraft and ships are required to notify AQIS if any passengers/crew are displaying symptoms of prohibited disease e.g. rabies, yellow fever, malaria, SARS, or avian influenza. Aircrafts are sprayed with insecticides. Airports also have mosquito-trapping programs to enable quick detection.

– Vaccination, including passive and active immunity

Vaccination primes the immune system to deal with a pathogen it has never been in contact with.

Passive/innate-> when antibodies are transferred to an unimmunised person, providing them with temporary protections against microbial agent or toxin

1. Natural: becomes immune due to antibodies from mother e.g. [placenta, breast milk](#)
2. Artificial: body is injected with antibodies from someone else e.g. [purified blood products of immune people or animals](#).

Active/Acquired-> occurs when immune system is activated by an antigen to make its own B, T cells and antibodies

1. Natural: becomes immune after catching a disease
2. Artificial: vaccination

– Public health campaigns

- Provide education and government regulations
- BreastScreen, Bowel Cancer, Cancer Council (Slip, Slop, Slap)

- Allow for early detection of potential diseases

– **Use of pesticides**

Pesticides are a broad term for insecticides, fungicides and herbicides.

The DDT was a widely used pesticide during WW1 that killed lice which transmitted the *Anopheles* mosquito which carried the plasmodium that causes malaria. However, its effectiveness was reduced as mosquitoes developed resistance. Now other insecticides such as pyrethrum are now used and are less harmful to the environment.

Another vector controlled by insecticides are aphids. Pesticides are sprayed widely to kill these vectors and control the spread of disease.

– **Genetic engineering**

Genetic engineering, through means such as recombinant DNA, allows for organisms to become resistant to diseases. Using this to produce disease-resistant plants/animals prevents the occurrence of disease in individual organisms, controls the spread of disease through the population and reduced incidence of the particular disease.

➤ Investigate and assess the effectiveness of pharmaceuticals as treatment strategies for the control of infectious disease, for example:

– **Antivirals**

Are used to control viral infections by inhibiting the development of viruses inside infected cells.

The viruses most commonly targeted by antiviral drugs include:

- HIV (HIV is the virus; AIDS is the disease caused by the virus)/
- Seasonal influenza A
- Herpes
- Hepatitis B and C

Antiviral medications used to treat influenza in AUS include:

- Oseltamivir (Tamiflu)
- Zanamivir (Relenza)
- Amantadine (Symmetrel)

– **Antibiotics**

Work by either killing or slowing down the growth of bacteria. Antibiotics are not effective against virus.

They are most effective when:

- Solely for treatment of bacterial infections not viral
- Bactericidal antibiotics are used to kill rather inhibit growth of the bacteria
- They are able to get to the site of infection and kill the bacteria
- Therapeutic blood levels are maintained.

Antibiotic resistance-> bacterial resistance limits the effectiveness of antibiotics in controlling outbreaks of infectious diseases. This problem rises as antibiotics become less effective over time in treating a particular bacterial disease e.g. MRSA (methicillin-resistant Staph) in hospitals is a major concern. Overuse in the 20th century has caused this problem.

➤ Investigate and evaluate environmental management and quarantine methods used to control an epidemic or pandemic

➤ Interpret data relating to the incidence and prevalence of infectious disease in populations, for example:

- **Mobility of individuals and the portion that are immune or immunised**

Pathogen features

- Growth-> how quickly and easily the pathogen replicates
- Survival-> How far the pathogen is able to travel and its opportunity to infect.
Adaptations cause increased chance of survival e.g. protect layer in protists, flagella
- Persistence-> allows them to replicate over a longer period of time and evade the hosts immune response e.g. lying low a dormant cell with slowed metabolic processes

Method of transmission

- Direct transmission e.g. sexual intercourse
- Indirect transmission e.g. vector
- Diseases are more likely to travel further via indirect transmission

Population density

- Diseases are spread quicker when people are in proximity, as it increases contact.
- High populations densities are linked to overcrowding and lack of infrastructure e.g. poor sanitation

Movement of individuals

- When infected individuals travel, they can spread disease to those they encounter.
People who have never encountered the disease before have no immunity, therefore increased likelihood of infection.

Proportion of the population that is immune

- Herd immunity-> if a significant proportion is immune to a disease, then any susceptible individuals (immunocompromised people e.g. HIV patients) are prevented from encountering the pathogen.

– **Malaria or Dengue Fever in Southeast Asia**

Dengue Fever

Pathogen and Cause	Caused by the dengue fever virus (DENV) which has five serotypes/strains, with genetic variations region specific. They are RNA viruses. Infection with one serotype gives lifelong immunity to that type but only short-term immunity to other types. Humans are the primary host of the virus, with the virus binding to WBC's and reproducing inside the WBC's which travel around the body.
Transmission	1. The bite of an infected Aedes mosquito 2. Infected blood products through organ donation
Symptom	Incubation period is usually 3-14 days after the bite. The patient is infected from the day before the fever, up to 12 days later. Symptoms include fever, extreme tiredness, headache especially behind the eyes, muscle and joint pain, nausea, loss of appetite, vomiting, diarrhoea, abdominal pain, red rash, itchiness, peeling of skin, hair loss and minor bleeding from nose and gums. Symptoms can last up to a week. In severe cases-> dengue shock syndrome with collapse from internal fluid, weak pulse, cold clammy skin and haemorrhage.
Host response	Viral infection of WBC's e.g. dendritic cells causes cells to produce cytokines and interferons which lead to many of the symptoms. Increased capillary permeability leads to fluid loss from blood vessels which causes low blood pressure and insufficient blood supply to vital organs. Infected dendritic cells move to lymph nodes and release mature viral particles by exocytosis that enter monocytes and macrophages, and the innate immune system is activated.

	B cells are sensitised to produce specific antibodies and sensitised T cells directly attack any infected cell.
Diagnosis	Initial diagnosis uses the appearance of symptoms and physical examination, low WBC count and positive tourniquet test. Confirmed by microbial lab tests
Treatment	Involves bed rest, maintaining fluid intake, avoiding mosquito bites to prevent transmission to other people, paracetamol for the pain and avoidance of anti-inflammatory drugs which increase chance of bleeding. In some cases, fluids are given intravenously and in severe cases blood transfusions are needed.
Prevention	Destroying breeding sites, wearing long-sleeved, light-coloured clothing, covering feet and using insect repellent. Mosquito coils and repellent devices. Biological control includes putting copepods and guppies in still water to eat the mosquito larvae.
Incidence	Was once limited to Southeast Asia however is now common worldwide, especially since WWII when the A. aegypti spread in cargo. Its global incidence is increasing being common in urban areas and a leading cause of illness and death in children in some Asian countries.

- Evaluate historical, culturally diverse and current strategies to predict and control the spread of disease

Middle ages- It was believed the body had four 'humours' - blood, yellow bile, black bile and phlegm which had to be balanced e.g. by bleeding out, using leeches, causing purging or vomiting. Many believed that disease such as the Black Death, was caused by horrible smells and that removing the smells e.g. by using a pomander they could repel the disease in the air. The pomander was a hollow container filled with spiced wax which people wore around their necks. They would also wear a beak mask filled with herbs and spices to keep away bad smells.

Traditional Chinese strategies- involves many strategies e.g. acupuncture, herbal medicine (ginseng, honeysuckle), maintaining a specific diet, massage and tai chi and qigong practices. Ancient Chinese doctors had noticed the patterns of some diseases with 'warm' diseases which went from one person in a household to another then from street to street and 'cold damage' diseases.

Current strategies- The Australian government has a National Framework for Communicable Disease Control was developed with the assistance of the states and territories and is available on the government website.

A reliable surveillance system is essential for the early detection of an outbreak of an infectious disease which requires practitioners to register the case with National Notifiable Diseases

Surveillance System (NNDSS). Once diagnosed and confirmed the case is investigated and the situation contained as much as possible to prevent further transmission.

- Investigate the contemporary application of Aboriginal protocols in the development of particular medicines and biological materials in Australia and how recognition and protection of Indigenous cultural and intellectual property is important, for example:

- **Bush medicine**

- ✧ Refers to the use of herbs and other plants to treat illnesses.
- ✧ Traditional Aboriginal remedies vary from region to region across Australia.
- ✧ Different treatments were needed for different conditions e.g. bites from snakes, spiders or centipedes, stings from jellyfish, wasps and scorpions, cuts and bruises, upset stomachs, heat stroke and eye and ear infections.
- ✧ Remedies did not remain static over time as trading and passing of information would have occurred.
- ✧ Some bush medicines related to introduced species e.g. billygoat weed a tropical American plant that is used as an analgesic and antibacterial external poultice over a wound.
- ✧ White kaolin clay was used to fix diarrhoea and stomach aches. It is now used in cosmetics and in medications that protect the epithelial lining of the gastrointestinal tract.
- ✧ Many people apply traditional and holistic healing in conjunction with modern medicine.
- ✧ Intellectual property-> property of the mind or privately owned knowledge. It is important when working with indigenous Australians and recording traditional ways and medicines that intellectual property is properly acknowledged and protected.

- **Smoke bush in Western Australia**

- ✧ *Conospermum* is endemic to Australia with more than 50 species.
- ✧ Many are adapted to bushfires and variable water supplies, being obligate seeders which involves the plant dying e.g. in a bush fire and stored seeds sprouting to form the next generation
- ✧ Intellectual property rights- In 1990's a biotechnology research and development country were awarded the right to smoke bush plant and related species to develop anti-AIDS drugs. The company paid over \$1.5 million based on projected royalties of \$100 million per year by 2002. In these agreements indigenous Australians were not acknowledge in any way. Concern for indigenous people's rights to intellectual property led to an investigation, recommending amendments to the Copyright Act and the formulation of new legislative arrangements.
- ✧ In 1993, after a meeting in north-eastern Australia of indigenous and non-indigenous specialists, the Julayinbul Statement on Indigenous Intellectual Property Rights was declared and insisted that Aboriginal common laws and Aboriginal intellectual property be acknowledged. The declaration also included the control and use of the genetic make-up of flora and fauna.

MODULE 8: NON-INFECTIOUS DISEASE AND DISORDERS



Homeostasis

Inquiry question: How is an organism's internal environment maintained in response to a changing external environment?

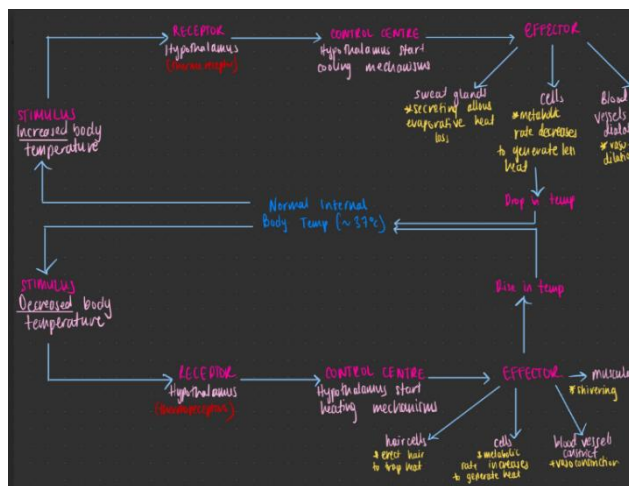
- construct and interpret negative feedback loops that show homeostasis by using a range of sources, including but not limited to:

Homeostasis -> maintenance of a relatively constant internal state, regardless of changes in the external environment. This is needed for metabolic functioning, removal of wastes and maintaining osmolarity.

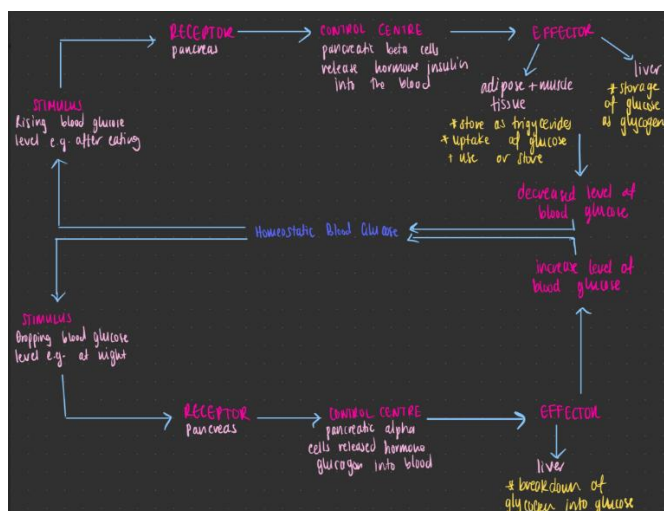
Negative feedback loops -> counteracting the change in homeostasis that is detected

- receptors detect a change in environment (stimulus)
- control centre receives information, determines a response and sends information where needed
- effectors bring about a change to counteract
- the control centre will also require feedback to stop the change

- temperature



- glucose



- investigate the various mechanisms used by organisms to maintain their internal environment within tolerance limits, including:

- **trends and patterns in behavioural, structural and physiological adaptations in endotherms that assist in maintaining homeostasis**

Adaptations:

Structural-> physical characteristics of an organism to enable its survival

Behavioural-> the way an organism acts to enable its survival (whole body)

Physiological-> the way the organism's body functions to enable its survival (part of the body)

- **Structural:** Camels have long eyelashes and closing nostrils to keep sand out of their eyes and nose, as well as a hump storing fat for energy and water.
Echidnas also have a dense coat of protective spines made of modified hairs for defense against predators.
- **Behavioural:** Wolves hunt in packs which increases their chances of successfully catching prey.
Birds and butterflies travel long distances to warmer regions during winter to find food and avoid harsh conditions.
- **Physiological:** snake producing venom to immobilise or kill prey and deter predators.

→ Water balance

- structural: Koalas have a large cecum, a pouch in their intestine where fermentation of eucalyptus leaves takes place. This cecum helps extract every last bit of moisture and nutrients from their tough diet.
- behavioural: Emus are most active in the cooler mornings and evenings. This reduces exposure to the hottest part of the day, minimizing water loss through sweating and respiration.
- physiological: Australian sea lion have metabolic water production: When breaking down fats and proteins for energy, sea lions produce metabolic water as a byproduct. This internal water source helps to offset water loss through breathing and sweating.

→ temperature regulation

- structural: Koalas have a thick coat of fur that provides some insulation. This fur helps to trap a layer of air around their bodies, which can act as a barrier to both heat and cold.
- behavioural: Emus are adept at finding and utilizing shade during the hottest parts of the day.
- physiological: Sea lions have a thick layer of blubber acts as a natural wetsuit. This blubber insulates their bodies in cold water, preventing heat loss.

- **internal coordination systems that allow homeostasis to be maintained, including hormones and neural pathways**

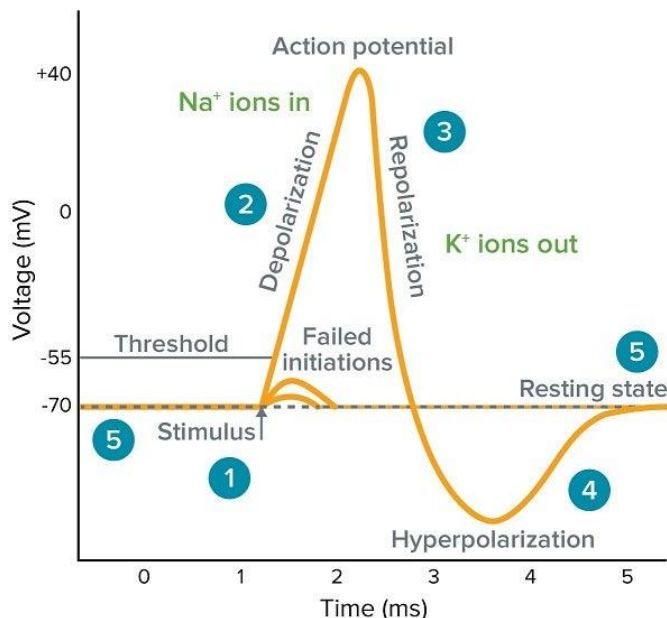
Hormones

- *Endocrine system*: glands secrete hormones and chemical messengers that travel in the blood. These hormones cause cells to change their activity, enabling them to maintain homeostasis e.g. melatonin-> hormone brain produces in response to darkness, Insulin-> hormone produced to maintain blood glucose level
- *Pituitary gland- master gland*: works with hypothalamus to release hormones to regulate other glands.
- *Thyroid gland*: releases thyroxine controlled by pituitary, hypothalamus, which increases metabolic rate.
- *Parathyroid gland*: small gland on the thyroid gland, secreting parathyroid hormone which controls calcium

- **Pancreas:** Islets of Langerhans have alpha cells (controlling release of glucagon) and beta cells (controlling release of insulin) which have chemoreceptors to release hormones.

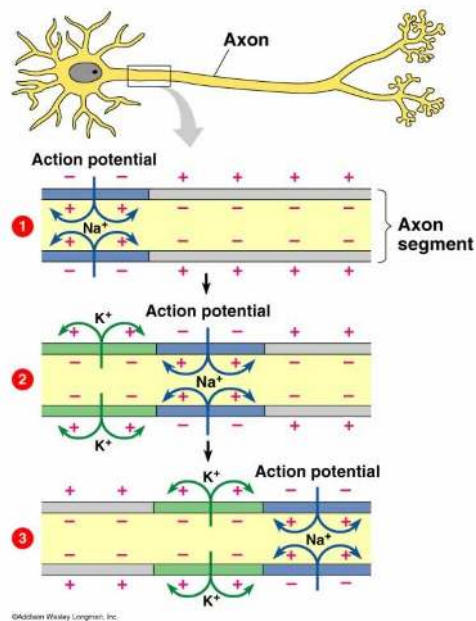
Neural pathways

- **Nervous system:** communication pathway made of 2 parts:
 - Central nervous system (CNS)
 - > brain: main control centre. Hypothalamus located centrally, links the neurons and endocrine system to control homeostasis e.g. body temperature, heart rate, blood pressure
 - > spinal cord: start from medulla oblongata (end of brain) and extends to vertebral column to waist. Contains nerve fibres linking the CNS and PNS conducting pathways to and from the brain. Also coordinates reflex action.
 - Peripheral nervous system (PNS)-> all other nerves; automatic nervous system, somatic nervous system (sympathetic and parasympathetic nervous system)
- **Receptors:** carry information as electrical impulses
 - Interreceptor- detect change internally (homeostasis)
 - Thermoreceptor- detect change in temperature (in skin and hypothalamus)
 - Chemoreceptors- detecting levels of chemicals in body (e.g. O₂, Co₂, pH)
 - Osmoreceptors- detecting changes in osmotic pressure in blood (found in brain)
- **Neurons:** nerve cells
 - 3 main parts
 - Cell body-> contains nucleus, organelles
 - Dendrites-> extensions of the cytoplasm to receive messages
 - Axons-> one long extension that transmits and sends impulses
- **Action potentials**
 - Membrane potential- difference of the electrical potential between the inside and the outside of the cell.



Na⁺ influx-> **depolarisation**

Depolarisation is when the difference in membrane potential is reduced, so it's becoming less negative (more positive). It starts before reaching threshold and continues after- it's not an isolated point but a process. During depolarisation, sodium is entering passively through the sodium channels. At +30mV the sodium channels close.



K⁺ efflux-> repolarisation

Repolarisation occurs at +30mV, when the potassium channels open and potassium leave the cell, bringing membrane potential to resting potential of -70mV. However, it can go even lower, therefore potassium channels close. The sodium potassium pump clears out excess potassium to thus reach resting potential.

The sequential opening and closing of Na⁺ and K⁺ channels ensure the action potential moves in one direction along the axon.

– mechanisms in plants that allow water balance to be maintained

Dry environment: Xerophytes (e.g. eucalyptus), mesophytes (e.g. roses)	○ <u>closed stomata</u>: decreases the amount of water on the guard cells so that they close the gap. Controlled by hormone <i>abscisic acid</i>, they close their stomata around midday and leave it open at night. ○ <u>Waxy cuticle</u>: providing a waterproof barrier, decreasing water lost. The thicker cuticle, the greater reduced water loss. ○ <u>Vacuole</u>: store large amount of water ○ <u>Reduced stomata</u>: less potential for water vapour to evaporate out of the plant ○ <u>Drooping leaves/ hairy leaves</u>: trap air around the stomata. Air becomes saturated with water vapour, creating humid microclimate where water isn't pulled out of the plant easily.
Wet environment: Hydrophytes (e.g. waterlilies)	● <u>Increased stomata</u>: help optimise gas exchange ● <u>Large SA:V ratio</u>: promotes water loss

Causes and Effects

Inquiry question: Do non-infectious diseases cause more death than infectious diseases?

➤ investigate the causes and effects of non-infectious diseases in humans, including but not limited to:

– **genetic diseases**

Caused by mutations of genes or chromosomes in an individual. Production of enzymes is coded for by genes which catalyse metabolic processes. If one enzyme is coded for by a mutated gene, then that enzyme will not be manufactured correctly and the metabolic pathway may not proceed as it should, causing genetic disease.

Examples of diseases caused by gene mutations include:

- Cystic fibrosis
- Albinism
- Huntington's Chorea
- Thalassaemia
- Sickle cell anaemia
- Phenylketonuria
- Haemophilia

Disease	Cause	Symptoms
Phenylketonuria	Cause by mutation of gene on chromosome 12 that codes for the enzyme phenylalanine hydroxylase (DAH) used to break down the amino acid phenylalanine (phe)	High levels of phe in blood and tissues will result in intellectual disabilities, delays in development, social, emotional, behavioural problems, psychiatric disorders and risk of seizures. Affects 1: 10000 babies in NSW. If levels are kept low, the babies will be normal. Adults who consume too much phe can have seizures or lapse into a coma. Need a diet low in protein and avoid sugar free products.
Cystic fibrosis	Caused by mutation of the cystic fibrosis transmembrane conductance regulator (CFTR) gene on chromosome 7. The CFTR gene is recessive.	Causes the mucus secreting organs to produce abnormally thick mucus and block passageways in lungs and digestive tract resulting in breathing difficulties, frequent lung infection, malabsorption of nutrients, passing of loud, foul-smelling motions, excess salt in sweat, liver failure, infertility and diabetes. One of the most common genetic disorders 1:2500 babies in AUS. Life expectancy is 30-40 years.
Down syndrome	Cause by mutation that results in the presence of a third copy	Results in growth delays, characteristic facial features, and mild to moderate intellectual

	of chromosome 21 (trisomy) in every cell.	disabilities, short stature, heart defects, and infertility. Down syndrome is not easily inherited, mainly due to the low fertility of those with the syndrome.
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– **diseases caused by environmental exposure**

Environmental diseases (ENVDS) are a non-communicable disease that results when people are exposed to toxic environmental chemicals. Other contributing causes of ENVDS include radiation, pathogens, allergens, psychological stress and others. WHO estimates that 24% of global disease is caused by environmental exposure.

Causes of environmental illness include:

- chemicals in cigarettes-> lung cancer
- exposure to asbestos-> cause tumours, lung cancer e.c.t.
- wood burning-> breathing problems
- unsafe drinking water-> cause allergens, cancer
- chemicals in workplace-> sterility in males and females
- Lead poisoning-> can cause high blood pressure, brain damage and stomach and kidney problems

Disease	Cause	Symptoms
Atherosclerosis	Poor diet, lack of exercise, excessive alcohol consumption. Smoking, high levels of stress	Hardened plaque builds up in artery walls, blocking blood flow to vital organs. Blockage can lead to cardiac arrest
Skin Cancer	Exposure to UV radiation from sun or solariums, the high energy radiation causes two thymine bases in DNA to bond together, altering the proteins produced. Causes frameshift mutation.	Tumours, excessive cell growth
Minamata Disease	Mercury poisoning. Minamata city in Japan was subjected to an environmental disaster in the mid-20 th century when a chemical company discharged mercury into the soil.	Is a neurological disorder causing lack of muscular coordination, physical and mental defects.

– **nutritional diseases**

Results from a lack of, imbalance, or over consumptions of certain nutrients

Disease	Cause	Symptoms
Scurvy	caused by lack of Vitamin C.	Causes swollen, spongy and purplish gums that are prone to bleeding, loose teeth, bulging eyes (proptosis), bleeding into the skin (severe and easy bruising), scaly, dry and brownish skin.
Type 2 diabetes	Diabetes results in reduced insulin, which is a hormone that	Formerly adult-onset diabetes. The risk of development increases with

	allows glucose to enter cells. Without glucose, cellular respiration cannot occur.	obesity, poor diet and lack of exercise, as well as genetic risk factors.
Rickets	Lack of vitamin D in children. Vitamin D is used to absorb calcium.	Results in pain in the bones, stunted growth, short stature, teeth deformities, bowlegs
Hypertension and Coronary heart disease	Cause by high fat diet, high cholesterol.	Involved build up of plaque in coronary arteries which supply blood to heart. Arteries become less elastic and harden (arteriosclerosis)
Obesity	Excessive intake of food, consuming more kilograms then expended.	Adverse effects such as increased blood pressure, atherosclerosis, stroke, type 2 diabetes and risk for cancer.

– cancer

Disease of the cell. Occurs when abnormal cells divide in an uncontrolled way, invading body tissue near them, spreading to other tissue and disrupting the normal functioning of the body.

Cell division is controlled and regulated by a particular set of genes contained in the DNA. These genes code for proteins that control the correct functioning of the process of cell division.

There are 3 types:

- **DNA repair genes:** code for proteins that are responsible for stopping the cell cycle while other proteins remove damaged regions of DNA and replace them with the correct sequence. Hence, if these genes mutate, they wont functioning correctly, DNA will not be repaired and damaged DNA will be replicated.
- **Proto-oncogenes:** code for proteins that stimulate cell growth and mitosis. Mutations of this gene will lead to expression of oncogenes that would normally be silent. This cause uncontrolled production of cells and prevent cell death.
- **Tumour suppressor genes:** codes for proteins that slow down or stop cell growth and mitosis. They also induce cell death if there is an uncontrolled increase in cell numbers. Mutations can halt the production of protein that control cell division and death.
- If these genes are mutated:
 - It doesn't allow cells to differentiate- can't perform specialised functions
 - Causes formation of tumours, if malignant (invasive/spreading) cancer results.
- Types of Cancer
 - **Sarcoma-** forms in connective tissue or muscle (bone, blood vessels)
 - **Carcinoma-** forms in epithelial tissue (skin, tissue lining internal organs)
 - **Lymphoma and myeloma-** forms in the lymphatic system and immune system)
 - **Leukemia-** forms in bone marrow and other blood forming tissue
 - **CNS cancers-** form in the brain and spinal cord.
- Causes of cancer
 - Smoking
 - Excessive alcohol consumption
 - Lack of physical activity, being overweight or obese and following a diet that is high in saturated fats

- Exposure to radiation e.g. UV radiation
- Contracting particular types of viruses, such as HPV or Hep B or C e.g. HPV is associated with cervical cancer
- Exposure to certain chemicals
- Inherited mutated genes that predispose a person to developing certain types of cancer e.g. BRAC1 gene
BRAC1, tumour suppressor gene found on chromosome 17, is believed to be responsible for coding for proteins involved in the repair of the PTEN gene. Mutations of the BRAC1 gene put women at much higher risk of developing this form of breast cancer.

➤ collect and represent data to show the incidence, prevalence and mortality rates of non-infectious diseases, for example:

- nutritional diseases
- diseases caused by environmental exposure

Epidemiology

Inquiry question: Why are epidemiology studies used?

➤ analyse patterns of non-infectious diseases in populations, including their incidence and prevalence, including but not limited to:

Epidemiology is the study of the distribution and determinants of health-related states or events (including disease) and the application of this study in controlling diseases and health problems such as collecting information to analyse and identify patterns in populations:

- ➔ patterns trends
- ➔ whether certain groups/ demographics are more susceptible
- ➔ monitoring methods to prevent and treat disease
- ➔ also looks at future research/areas to focus on

Incidence-> number of new cases [shows changes, can show if prevention and control work]

Prevalence-> number of total cases

Mortality-> number of deaths due to disease

Epidemiological Studies

- Descriptive studies

Usually the first of study, providing information about the frequency of the disease, which section of the population is affected (age, gender, SES ect), geographic location and timeframe. Allows early hypotheses on cause of disease

- Analytical studies

Used to collect more data to statistically analyse hypotheses to find likely causes of disease. Uses mortality, morbidity, incidence and prevalence.

Divided into two types:

- ➔ *Cohort studies*-> studying two or more similar groups of people who are free of the disease and exposing one to the possible cause. They are then followed for some time to compare incidence
- ➔ *Case-control*-> comparing people with the disease (case) to people without (control) and compare difference to find cause

- Intervention studies

Testing the effectiveness of a treatment or public health campaign e.g. experiment study allowing testing of new drugs by giving one group the trial drug and the other placebo.
Testing effectiveness of medication and compare to what is already on the market

Evaluating epidemiological studies

Epidemiological studies should have:

- Conducted over a long period of time
 - Study a large sample size (thousands)
 - Collect relevant data from affected and unaffected people
 - Participants need to represent the wider population (e.g. age, gender, diet)
 - Controls should have similar aspects to the case cohort (e.g. similar ages if that isn't being tested)
 - Collect data on incidence, prevalence, mortality and morbidity -> use this for statistical analysis
 - Identify causes of disease and risk factors
- **nutritional diseases**
 - **diseases caused by environmental exposure**
- investigate the treatment/management, and possible future directions for further research, of a non-infectious disease using an example from one of the non-infectious diseases categories listed above

Melanoma

Treatment: common treatment is surgery-removing the tumor and skin around the tumour to reduce recurrence. More advanced cases require radiation, chemotherapy and targeted therapies depending on far the melanoma has progressed, and the patient.

Radiation-> exposing cells to radiation kills cancerous cells, normal cells repair. It is administered externally or by placing radioactive source near cancer cells.

Chemotherapy-> can slow the growth of cancer cells (not so useful for melanoma, so used less)

Targeted therapy-> taking drugs that affect molecules that aloe the growth of tumour cells e.g. disrupting pathways for mitosis/cell division)

Immunotherapy-> using vaccines with antigens for melanoma or using drugs that better aloe the immune system to recognise and kill cancer cells (checkpoint inhibitors)

- evaluate the method used in an example of an epidemiological study
- evaluate, using examples, the benefits of engaging in an epidemiological study

Prevention

Inquiry question: How can non-infectious disease be prevented?

- use secondary sources to evaluate the effectiveness of current disease-prevention methods and develop strategies for the prevention of a non-infectious disease, including but not limited to:
- Prevention>managing a disease
 - Some diseases have no treatment, some treatments have side effects, therefore, prevention is integral
 - Treatment can create financial burden
 - Strategies to prevent one important to improve the overall health and wellbeing of public. Strategies use **epidemiological data**.
- **educational programs and campaigns**
 - provides information and educate population about effects of a disease and its risk factors (and therefore protective behaviours)
 - needs an evidence-base for action, evidence-based intervention, and real time monitoring, evaluation and program improvements.

Case Studies

1. Slip, slop, slap, seek, slide (for skin cancer e.g. melanoma)

- Developed in Victoria (1980s) to reduce incidence of skin cancers and change sun exposure behaviour.
- Involved television advertisement, sunscreen vans, catch phrase, jingle/song, free bottles on sunscreen and more gov. funded research.
- Led to a decline in melanoma, and though to have prevented 43,000 cancers.

2. QUIT campaign (for smoking)

- Smoking is leading preventable cause of death in Australia population with 15,500 smoking-related death each year.
- QUIT was developed to reduce prevalence of smoking-related disease in the population
- Lung cancer, heart disease, lung disease and other cancers has clear links with smoking -> the longer a person smokes, the greater their chances of developing disease

QUIT has a multifaceted approach and include strategies such as:

- Use of slogans
- Graphic images on tobacco packets to show dangers of smoking
- National helpline
- Legislation -> requires plain packaging
- Increased tax
- Legislation to restrict advertising
- Banning smoking in certain areas (smoking-free zones)

Has been evolving and improving over the last 44 years-> 1991 was a 24% smoking rate, 2022 decreased to 8.2%.

- genetic engineering

- pre-implantation genetic testing (IPG) to prevent genetic diseases
 - ➔ parents undergoing genetic counselling to determine if offspring will inherit the condition or using IVF to ensure offspring are born free of the disease
- transgenic species to prevent nutritional diseases
 - ➔ vitamins deficiencies and poor diet are issue in developing countries; transgenic species can supplement certain nutrient (e.g. Golden rice)
- Can be used to develop vaccines (e.g. Gardasil for HPV, which causes cervical cancer)
- Used as a screening tool for disease susceptibility
 - ➔ Prediction of predisposition or susceptibility to disease, which is essential for prevention-> improved screening, diagnosis and early intervention to prevent non-infectious disease.

Technology and Disorders

Inquiry question: How can non-infectious disease be prevented?

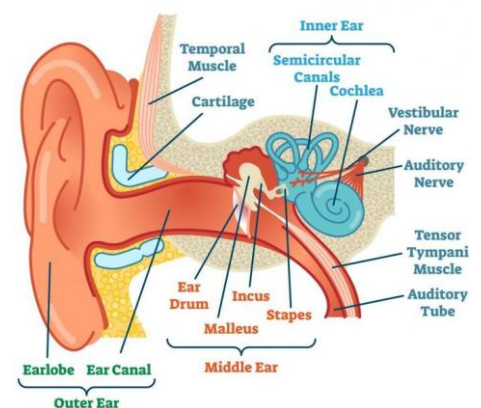
- explain a range of causes of disorders by investigating the structures and functions of the relevant organs, for example:

– **hearing loss**

Structure and function of the ear

Pinna	Collects and directs sound waves into the ear canal
Auditory canal	Channels sound waves toward the tympanic membrane (eardrum) and protects the middle ear
Eardrum (tympanic membrane)	Vibrates in response to sound waves, converting sound energy into mechanical vibrations
Ossicles [hammer (malleus), anvil (incus) and stirrup (stapes)]	Amplify and transmit vibrations from the eardrum to the oval window of the cochlea
Oval window	Transfers vibrations from the stapes to the fluid inside the cochlea
Cochlea	Converts mechanical vibrations into electrochemical impulses; hair cells in the organ of Corti detect sound frequencies
Organ of Corti	Contain sensory hair cells that bend in response to fluid movement, generating nerve impulses
Auditory nerve	Carries electrical impulses from the cochlea to the auditory cortex of the brain for sound interpretation.
Eustachian tube	Equalise air pressure between the middle ear and the external environment to ensure proper vibration of the eardrum
Semicircular canals	Detect rotational movement and help maintain balance (not involved in hearing directly)

1. Sound waves (sound energy) are directed into the **outer ear canal** by the **pinna** and set up **vibrations (kinetic energy)** of identical frequency in the eardrum (**tympanic membrane**)
2. These vibrations are transferred to the **ossi cles- the hammer, anvil and stirrup**. These bones **amplify and transfer kinetic energy to the oval window**, which is the membrane that separates the middle ear from the fluid filled inner ear.
3. Vibrations **pass from the oval window into the upper canal**, passing the middle canal through to the lower canal to the round window which acts as a valve, **relieving built up pressure from the sound waves entering into the cochlea**. The **Cochlea** is a fluid filled organ with 3 canals (upper, middle, lower) and contains *the Organ of Corti*.
4. The middle canal contains receptor cells (hair cells) located between the **basilar** membrane and **tectorial** membrane



5. When sound waves move through the canals, **it pushes up on the basilar membrane**. This causes **bending of the hair cells** against the tectorial membrane.
6. Bending of the hair cells **stimulates electrical impulses which are transferred to the brain** by the **auditory nerves**. (converts kinetic energy to electrochemical energy)

Types of hearing loss

Conductive hearing loss-> caused by a problem with the mechanical conduction of vibrations through the outer and middle ear. There are numerous causes including: malformation of the structures in these regions, a perforated eardrum, infection in the outer/middle ear, damage to the ossicles due to trauma such as exposure to sudden loud noises or hardening of the stapes bone.

These factors inhibit the movement of vibrations through the outer and middle ear. It is the loudness of the sound that is usually affected.

Sensorineural hearing loss-> caused by damage to, or malformation of, the inner ear, including parts of the cochlea, hair cells or auditory nerve. The causes include: excessive noise exposure, heredity, birth defects, infections, tumours, medication and ageing. Damage to the inner ear (including the receptor hair cells) prevents the kinetic energy of the vibrations being transformed into electrical impulses to be sent to the brain for interpretation.

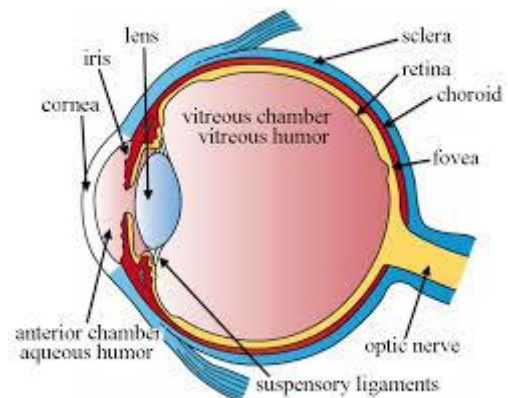
– visual disorders

Structure and function of the eye

The eye is used to detect light in to give us vision. The eye is made up of 2 types of photoreceptors:

- 1) **Rods**: used in low light conditions, detect blue/grey
- 2) **Cones**: used in well-lit conditions; three types (red, blue and green)

The eye refracts (bends) light and focuses it onto the retina where the photoreceptors are located. A message is then sent to the brain to interpret the light as an image.



Conjunctiva	○ Membrane that covers the surface of the eye and inside of eyelids. ○ Protects front of eye
Sclera	○ Tough, thick, white part of the eye ○ Protects the eye and helps maintain
Cornea	○ Front part of the eye. Clear part of the sclera ○ Protects the front of the eye and bends light rays they pass through it
Choroid	○ Layer inside the sclera, thin black layer containing blood vessels ○ Absorbs any stray light preventing false images
Retina	○ Innermost layer of eye, it lines the back of the eyeball and is covered in light sensitive photoreceptors and nerves
Iris	○ Coloured part at the front of the eye, it is composed of muscles ○ Expands and contract to control the amount of light entering the eye
Pupil	○ Opening in the centre of the iris, its size is controlled by the iris

	○ Allows light to enter the eye
Lens	○ Transparent biconvex () disc behind the pupil ○ Focus the light entering the eye
Aqueous humor	○ Viscous liquid that fills the front chamber of the eye ○ Helps keep the eyeball in shape, also bends light passing through it
Vitreous humor	○ Jelly-like liquid which fills the large back chamber of the eye ○ Helps keep the eyeball in shape, also bends light passing through it
Ciliary body	○ Connects the choroid with the lens, it contains suspensory ligaments and ciliary muscles ○ Ligaments hold the lens in place; muscles alter the shape of the lens
Fovea	○ Also called the 'yellow spot' area on the retina directly in line with the pupil and lens ○ This is the area with the highest concentration of photoreceptors. It is the 'sweet spot' of the retina
Optic nerve	○ Connects the eye to the brain. The area where the optic nerve enters the eye is called the 'blind spot' because it has no photoreceptors, images cannot be produced.

Visual disorders

Myopia and Hyperopia

- Can result when:
 - > eyeball is not correctly shaped
 - > ciliary muscles are weak
 - > lens is not flexible enough
- This causes the eye not to be able to focus light onto the retina properly, resulting in a blurred image

Myopia: If your eyeball is long or lens too rounded, you can see close objects but not distant ones (Short-sightedness). The image falls before the retina.

Hyperopia: If your eyeball is short, or lens too elongated you can see distant objects clearly but not close ones (Far-sightedness). The image falls behind the retina.

Cataracts

- The lens is a living structure even though it has no direct blood supply.
- A cataract is any loss of transparency in the lens. This results in becoming cloudy leading to a decrease in vision and total blindness if not treated. Having cataracts can be like looking through a cloudy window.
- Cataracts can be caused in several ways:
 - > Ageing: senile cataracts
 - > Present at birth: congenital cataracts
 - > Developing from disease such as diabetes
 - > Excessive use of steroids
 - > Injury such as trauma or exposure to infrared or radiation
- Loss of sight from cataracts can usually be treated by removal of the lens and implanting another.

Muscular Degeneration (aka aged-related macular degeneration)

- Progressive disease that causes degeneration of cells beneath the retina (macula)
- Affects central vision which impacts ability to read, recognise faces, drive ect.
- Risk factors include:
 - > Smoking
 - > Excessive exposure to UV radiation

– loss of kidney function

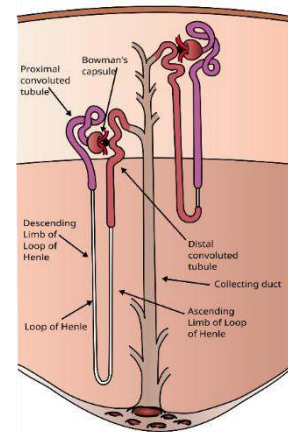
The kidneys are a part of the excretory system. They have two main functions 1) remove nitrogenous wastes from the blood 2) Maintain optimal salt and water levels (osmoregulation)

Structure and function

- Has 3 main functions: filtration, reabsorption and secretion.
- Has 3 main regions: cortex, medulla and pelvis.
- The functional unit of kidneys is the nephron. Each kidney contains millions of these units, extending from the cortex into the medulla.

Nephron structure\

Nephron-> each nephron consists of a Bowman's capsule, which is spherical, 2-sided hollow, cuplike structures that contains a spherical network of capillaries called the glomerulus. Far side of the Bowmans capsule leads into the proximal tubule, the loop of Henle and the distal tubule, which leads into the collecting duct.

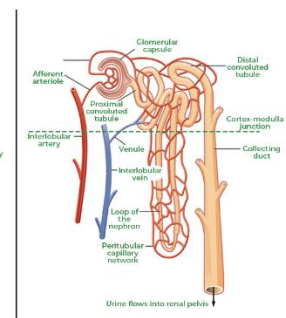
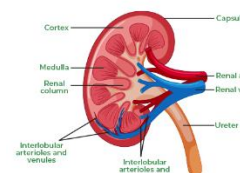


Bowman's capsule	When blood flows through the glomerulus, the waste products such as nitrogenous waste products including amino acids and glucose, pass through the glomerulus and into the bowman's capsule. This fluid is called glomerular filtrate and carries into the tubules.
Proximal tubule	Carries the glomerular filtrate into the loop of Henle. Water and sodium chloride passively move through the tubule into interstitial fluid.
Loop of Henle	A u-shaped segment that is carried from the proximal tubule. Its thin descending limb is permeable to water, passively, and the ascending limb is impermeable to water, but permeable to salt, passively in the inner medulla and then actively in the outer medulla. It is followed by the distal tubule
Distal tubule	Carries the fluid out of the Loop of Henle. Reabsorption of water occurs.
Collecting duct	Reabsorption of water occurs. Carries fluids to renal pelvis

Filtration: Under high pressure, blood travels from the heart via the aorta to the renal artery and into the kidney.

The renal artery branches into smaller vessels until millions of capillaries are formed. Each capillary enters a nephron, where it forms the glomerulus inside the hollow of each Bowmans capsule.

Filtration is based on size of substances. Substances that are small enough will pass through the walls of



the glomerulus into the Bowman's capsule such as nitrogenous waste products and water with dissolved amino acids, glucose and salts (ions).

Once inside the Bowman's capsule, this fluid is called the glomerular filtrate.

Reabsorption: Returns essential components that have been filtered out of the blood back into the bloodstream. Such as

- Glucose
- Amino acids
- Some vitamins,
- Minerals
- Hydrogen bicarbonate
- Ions (Na⁺, K⁺)
- Water

All solutes that are reabsorbed from the nephrons move by active transport and facilitated diffusion.

Glomerular filtrate also contains relatively high concentration of dissolved urea and other wasters, most which are not reabsorbed.

Secretion: third process that leads to urine formation. The kidneys are continuously processing an enormous amount of blood (~125ml of blood per min) to form a small amount of urine.

Involves removal of toxic substances including:

- Urea (via diffusion)
- Uric acid
- Ammonia (via diffusion)
- Hydrogen ions
- Drugs such as penicillin, saccharin and morphine

Antidiuretic hormone (ADH) and Aldosterone-> salt and water are reabsorbed from the nephron tubules into the bloodstream depending on the bodies needs.

The amount of salt and water reabsorbed is controlled by two hormones 1) Antidiuretic hormone (ADH or vasopressin) 2) Aldosterone

Antidiuretic hormone (ADH): also called vasopressin, it controls/stimulates the reabsorption of water by adjusting the permeability of the collected ducts and distal tubules. ADH is made in the hypothalamus but stored/released by the pituitary. Receptors in the hypothalamus monitor and control the concentration of blood.

If the body has...

High salt concentration	○ ADH levels increase ○ Collecting ducts and tubules become more permeable to water ○ More water reabsorbed ○ Concentration return to normal (concentrated urine)
Low salt concentration	○ Less ADH released ○ Collecting ducts and distal tubules less permeable to water ○ Less water reabsorbed ○ More water passes out with urine ○ Concentration return to normal

Aldosterone: Produced and released by the adrenal glands which sit above the kidneys. Controls the amount of salt in the blood by stimulating the reabsorption of salt in the loop of Henle to regulate salt and water balance in the kidney.

If the body has...

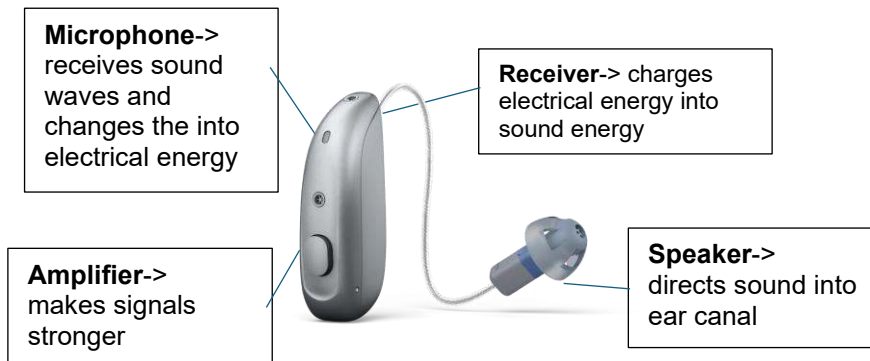
High salt levels	○ High blood volume and pressure due to large amounts of water diffusing back into blood. ○ Aldosterone levels decreased ○ Less salt reabsorbed, therefore, less water diffusing into blood ○ Salt levels decreased, blood volume and pressure decreases
Low salt levels	○ Low blood volume and pressure due to water diffusing out of blood ○ Levels of aldosterone increased ○ More salt reabsorbed, therefore, more water diffusing into blood ○ Salt levels increased, blood volume and pressure increases

Loss of kidney function

- Usually occurs gradually
 - Usually involves damage to the nephrons. This prevents the nephrons from carrying out its function properly: filtering, reabsorption and secretion
 - Resulting in wastes not being removed efficiently and an inability to balance water and salt levels in blood.
 - Can be caused by:
 - Type 1 and 2 diabetes
 - Continued high bp
 - Recurring kidney infections and prolonged use of certain medications/drugs
 - Kidney stones, tumours
 - Damage to kidneys (injury/drugs)
 - Symptoms include:

➤ Nausea	➤ Fatigue/weakness
➤ Vomiting	➤ Changes in urine volume
➤ Sleep problems	➤ Hypertension
➤ Chest pain	➤ Shortness of breath
➤ Swelling of feet/ankles	➤ Persistent itching
➤ Loss of appetite	
- investigate technologies that are used to assist with the effects of a disorder, including but not limited to:
- evaluate the effectiveness of a technology that is used to manage and assist with the effects of a disorder
- hearing loss: cochlear implants, bone conduction implants, hearing aids

Hearing aids-> A battery-operated electronic device that fits into the hollow outside the ear canal. Works by amplifying sound waves within the ear canal.

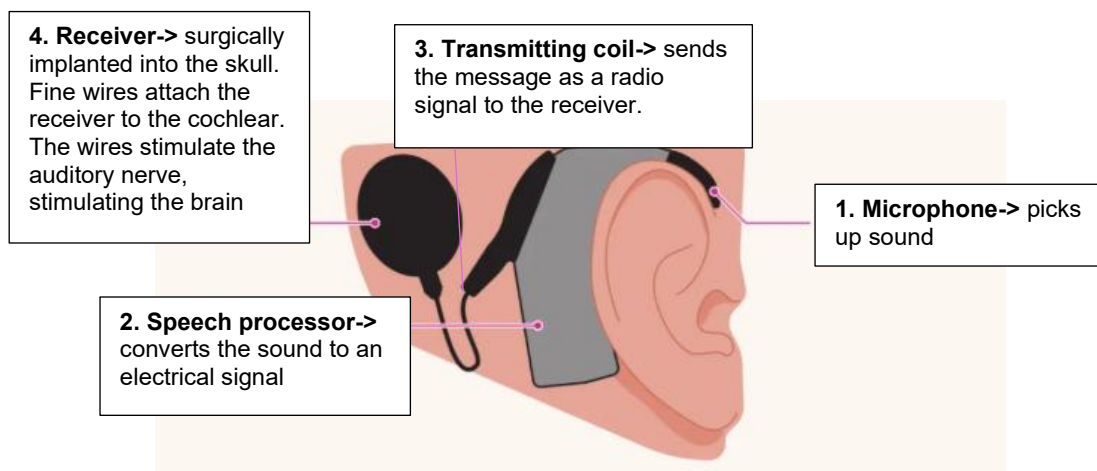


Eardrum, ossicles and cochlear must still have an adequate level of function so that sound can be registered and converted into an electrochemical sound. Can be used for people with conductive hearing loss.

Advantages	Disadvantages
○ relatively cheap ○ easy to install (no surgery) ○ non-invasive	○ amplifies all noises (background noises) -> painful ○ does not assist those with sensorineural hearing loss

Bone conduction implant-> creates sound vibrations that are conducted directly to the cochlear. Used for conductive hearing loss. Inner ear must be working.

Cochlear implant-> used by people with profound hearing loss- typically sensorineural (inner ear). Works by converting sound into electrical signals and directly stimulating the auditory nerve.



Advantages	Disadvantages
○ Provides hearing to profoundly deaf people	○ Highly expensive ○ Side effects (infection, drooping face numb tongue due to nerve damage)

	○ Wearers must learn to interpret sound ○ Success is limited
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– **visual disorders: spectacles, laser surgery**

Spectacles-> frames that hold corrective lenses made of clear, hard plastic. The shape is determined by the visual disorder that needs correction.

- **Myopia**: can be corrected wearing **concave** lenses which are thicker towards the outside and thinner towards the centre. These bend the light rays' outwards, causing them to diverge before they reach the eye.
- **Hyperopia**: can be corrected by wearing **convex** lenses which are thicker towards the centre and thinner towards the edges. This bends the light inwards, causing them to begin converging before they reach the eye.

LASIK/ Laser surgery->

1. A thin flap of the cornea is lifted up
2. A laser (high energy light beam) is used to reshape the cornea
3. For Myopia-> curvature of the cornea is decreased
For Hyperopia-> curvature of the cornea is increased
4. The flap of the cornea is repositioned and it rebounds to itself.

– **loss of kidney function: dialysis**

- People with abnormal functioning kidneys are not able to remove wastes such as urea. They have to undergo *renal dialysis* to regulate their blood.
- There are two forms of dialysis:
 1. **Haemodialysis**:
 - ➔ Blood is taken out of the body through a vein and passed into a dialyser, which is a bundle of hollow fibres (coiled to increase SA so rate of diffusion increases), made of semi-permeable membrane\.
 - ➔ The dialyser is submerged in dialysis fluid (dialysate) w hich flows in the opposite direction to the blood to maintain a concentration gradient to maximise diffusion. Dialysate has the same concentration as blood plasma, with the metabolic waste. Because the concentration of metabolic waste is higher in the blood than in the dialysing fluid, the waste materials move through the semi-permeable membrane into the dialysing fluid by diffusion.
 - ➔ An anti-clotting agent, *heparin*, is added to prevent clotting. The cleaned blood is then returned to the body. Continual replacement of the dialysate is required.
 - Disadvantages:
 - Renal dialysis must be done 3-4 hours, 2-4 times per week
 - Time consuming
 - Require external machine
 - Side effects-> low bp, vomiting, light-headedness
 - Only limited amount of waste can be removed from blood
 - Substances such as sodium phosphate and potassium ions do not diffuse quickly enough and therefore may accumulate in the blood. Patients have to follow a specific diet to prevent this.
 - Advantages:
 - Can occur in hospital or in home

- Can be done during sleep

2. Peritoneal dialysis

- ➔ Removes waste products from the blood.
- ➔ During peritoneal dialysis, a cleansing fluid flows through a tube into part of the stomach area (abdomen).
- ➔ The inner lining of the abdomen, known as the peritoneum, acts as a filter and removes wastes from blood.
- ➔ After a set amount of time, the fluid with the filtered waste flows out of the abdomen and is thrown away.

- Disadvantages

- Need to be able to set up and attach equipment
- Risks include hernias, weight gain, infection at catheter site, weakening of abdominal muscles

- Advantages

- Uninterrupted time, most patients able to continue working
- Can be done during sleep